

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

Section 17

Gradient system

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1. SETTING TO WORK THE GRADIENT SYSTEM

WARNING

While working on the gradient system be sure the AC power to the gradient system is off. Wait 5 minutes for the supply capacitors to discharge. Measure from ground to the high voltage bus, see drawing 'gradient amplifier cabinet lay out', to be sure the supply capacitors have discharge to less than 2 V.

BE AWARE THAT THERE IS 0.2 FARAD CAPACITY IN THE SYSTEM

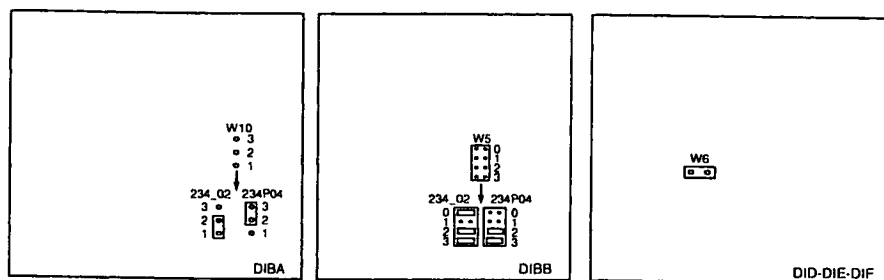
The gradient system is built in a standard 19 inch rack, at the top the interface unit (DI) with the WFG Waveform Generator WFG boards (Eddy current), followed by 3 power amplifiers model 234 and at the bottom a 15 kW 160V DC power supply. A separate power supply (28 volt DC) for the cooling fans of the power amplifiers is located behind the mains panel of the gradient cabinet, see drawings manual. The BDAS/Gradient cable together with the mains and temp. sensor cables are routed on the left side (seen from front) of the cabinet to the interface unit.

1.1. FIRST TIME POWERUP THE SYSTEM

Before applying power to the cabinet check that the system interconnections and the grounding straps from the side panels and doors are connected properly. The gradient 234 Copley amplifiers are supplied standard with mounted **NGC**, if not install the correct headers (delivered with system).

Check the jumper setting for the gradient interface boards:

- DIBA: (GRACRMB) W10 selects the supplied clock frequency :
1-2 for STANDARD (81 kHz for Copley234_02)
2-3 for OMNI (100kHz Copley 234P04)
- DIBB: (GRACRDB) W5 selects the amplifier configuration.
- DID, DIE, DIF: (WFG/ECC) W6 enables A0 and T0 (A0 and T0 are factory adjusted !)



FB01.DRW

Check with an ohmmeter the X channel output+ to the Y, Z channel output+ to be sure there is no continuity (the ohmmeter should read above 1 kOhm). Repeat this for the Y channel output+ to the Z channel output+. Typical value is 1.8 kOhm.

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Precondition:

- Check the settings of the gradient F1 breaker in the MDU. Settings must be $I_r = 32$ (yellow), $I_{rm} = 450$ (red). Refer to section 5.
- Verify the dipswitch settings of the 234_copley's, according following table.

CHANNEL	DIPSWITCH S901:5 SYNC	DIPSWITCH S901:6 CLOCK
X	OFF	ON
Y	OFF	ON
Z	OFF	ON

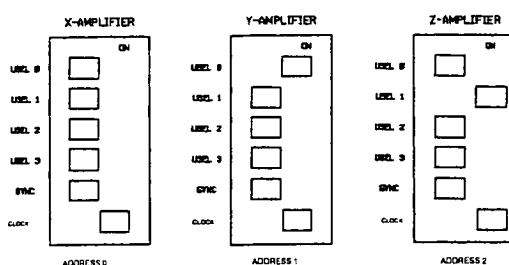


Figure 1: Dipswitch settings Copley-234

- The NTDAC must be ON, otherwise the sync_clock is missing.
- The gradient cooling ventilator should be ON and magnet covers mounted. When the gradient cooling ventilator is not yet operational or covers not yet mounted, overrule temporarily the gradient air-pressure sensor. Make a connection on the TFEI chassis connector MDI5X1: between pin 1 and pin 6.

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Perform the following procedure to switch on the Copley 234 gradient amplifier. If you run into problems while checking, resolve the problems before you continue with further testing.

1. Switch off the 15 kW power supply unit (DP) with 'POWER' switch.
2. Press the inhibit switches on all the amplifiers DM1, DM2 and DM3 (red button).
3. Switch on the gradient amplifier cabinet with switches marked as 'MAINS' and 'AUX' located at the bottom front panel.
4. The **green** 'power' led's of the WFG controller and WFG Interface rack boards are ON.
5. Check operation of all fans.
6. Switch on the 15 kW power supply, with the 'power' switch, the output voltage automatically goes to the low power mode (40 V). The green **power** on led and the yellow **voltage** led are ON.
7. On the amplifier modules the 'inhibit-' and 'normal' led must be **red** (all others LED's are OFF)
8. Press again the 'inhibit' switch on the X amplifier (red button inhibit led = OFF). The 'normal' and 'power' led's should light **green** and the others LED's OFF.
9. Repeat the previous procedure also for the Y and Z amplifier
10. The 'channel on' led is only ON (green) after down-loading SW or a restart.
If the system is not used for a few hours, the 'channel on' led goes OFF and will be activated after a start of a scan.

Verify Gradient amplifier parameters/drivers in ASW

```

Login :      GYROTOOL.
Enter:      <RETURN>.
Select:     Modify system configuration
Select:     Scanner hardware configuration
Select:     Gradient amplifier type
                        Copley_234_02 (STANDARD)
                        or
                        Copley_234P04 (OMNI)
Select:     <Proceed>
Select:     <Exit>
Select:     EXIT from GYROTOOL
  
```

If gradient amplifier type has been changed a GYRORESTART must be done.

Login under GYRORESTART

Enter: YES

Verify Gradient amplifier drivers in TSW:

```

Startup:     TSW
Select:     CONF
Select:     UPDATE
                The scanner hardware configuration is copied from the ASW configuration file.
                If this is finished, inspect the correctness of the TSW configuration file.
Select:     INSPECT
Verify:     Which gradient amplifier type is selected. The value must be 'Copley-234'.
Select:     QUIT
  
```

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Standard	Omni	Omni

2. MEASUREMENTS ON THE GRADIENT SYSTEM

2.1. GRADIENT AMPLIFIER CHECK/ADJUSTMENT PROCEDURE

Precondition:

- The BDAS and gradient amplifier must be allowed to warm up for at least 30 minutes.
- The RF Amplifier must be switch ON.
- The RF door must be closed.

Access to GYROSCAN gradient adjustment procedure:

- Login under GYROSCAN.
 - Select: <Scan Control>
 - Select: <Scan Utilities>
 - Select: Enter service mode <password>
 - Select: System tuning <password>
 - Select: System tuning tools
 - Select: Installation procedures
 - Select: Gradient adjustment procedures

2.2. CLOCK VERIFY TEST

Select: Clock verify test

An information screen is displayed (showing the jumper setting of W10 on DIBA in the gradient interface).

Select: <Proceed>

For STANDARD (Copley 234_02) the system should respond with **81000Hz**.

For OMNI (Copley 234P04) the system should respond with **100000Hz**.

If the result is not within specification, check the setting of jumper W10 and also the setting under figman (Gyrotool).

2.3. OFFSET CHECK (THIS IS NOT A SYSTEM TUNING TOOL)

Measure the 'current' monitor signal of each Copley amplifier (X,Y and Z) on the BNC connectors, with a DVM. The voltage must be < 50 mV.

During this check the gradient must not be pulsed (no demand signals).

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Standard	Omni	Omni

2.4. GRADIENT CURRENT ADJUSTMENT (COPLEY GAIN ADJUSTMENT)

This test generates simultaneous for all axes, a defined 240 A gradient pulse. The measured current values for all channels must be 240 A with a tolerance of ± 3.5 A. The three axis currents are measured in one test called 'Gradient current adjustment' and the results are displayed on the console monitor, see figure 2. The current is adjustable with the 'GAIN' potentiometer of the Copley 234_amplifier, see drawing 'connector location 234_amplifier' Z2-6.2. (Drawings Manual).

To zoom out a part of the displayed signal, drag the desired window by moving the mouse, with the left mouse button pressed, over signal part of interest, or just toggle the autoscale button (on and off again). By clicking the reset button, your original signal will return.

Precondition: The RF door should be closed.

Procedure:

Select: Gradient current adjustment

An information screen is displayed

Select: <Proceed>

To start the measurement

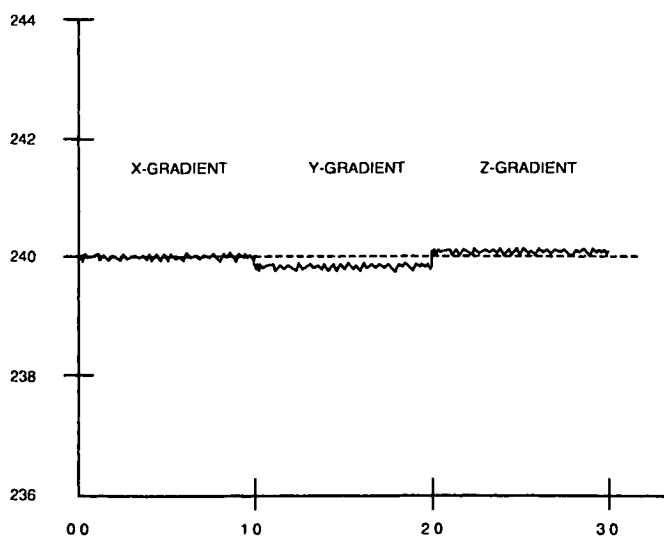


Figure 2: Gradient current adjustment, displayed on the console monitor

When adjustments are within the limits, preferable as close as possible to the 240 A value:

1. Enter: EXIT
A MRL will be displayed, check if all values are within the specifications (no error messages).
2. Select: <Proceed> To exit the MRL listing.
3. Select: <Cancel> To exit the 'gradient adjustment procedures' menu.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

3. EDDY CURRENT CHECK PROCEDURE

Precondition:

The gradient orientation must be correct. How to check this is described in paragraph 'Check of the gradient orientation' of section 16 of the manual (SMI).

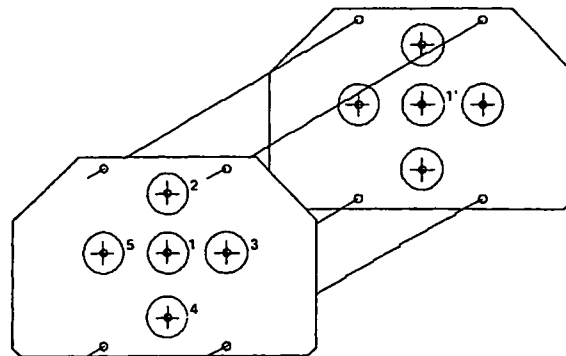
Required tools:

- Eddy current compensation set 8122 101 45004, which includes the Eddy current pick-up coil holder
- The eddy currents are adjusted on the WFG/ECC boards inside the Gradient Interface rack. Since the measured signal is displayed on the operator console, a temporary display must be organised, for external monitoring in the technical room, see the appendix of chapter 11.

NOTE

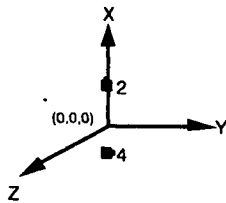
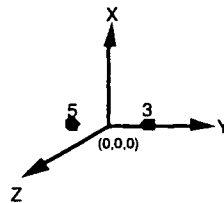
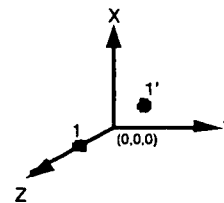
- *Take special care with positioning the fluxmeter, the drift is strongly dependent on environmental temperature variations.*
- *Keep the flux meter away from the gradient cabinet and gradient cables to eliminate the possibility of forced induction into the test equipment, the minimum distance > 1 meter.*
- *Use the coax cables, as delivered with the set, for the connection between the flux meter and the Gradient interface rack DIB to minimize the noise:*
 - ☐ The 'thick' coax cables for the signals;
 - ☐ The 'thin' coax cable for the 'reset' signal.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
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Figure 3: Position Eddy current pick-up coils

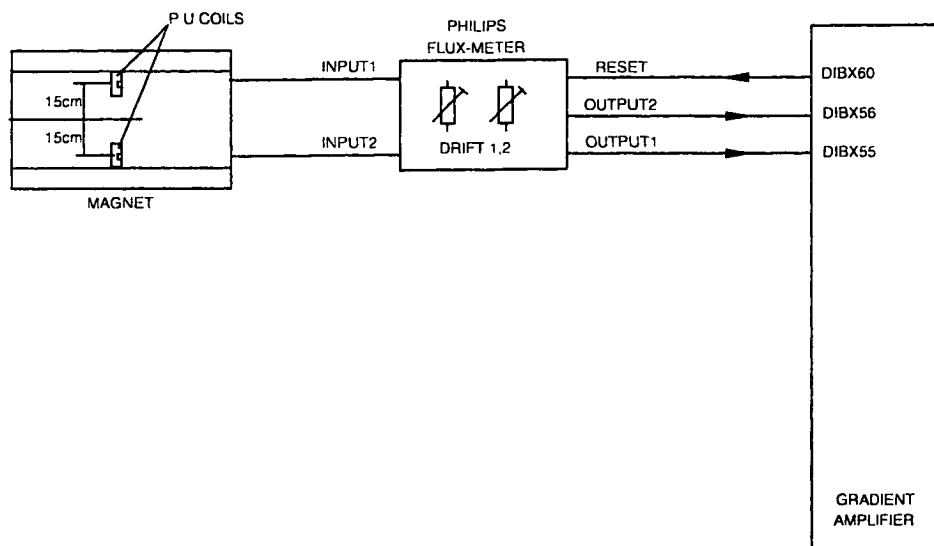
SET UP FOR FLUX DRIFT AND
X-WFG BOARD ADJUSTMENTSSET UP
Y-WFG BOARD ADJUSTMENTSSET UP
Z-WFG BOARD ADJUSTMENTS

JO19A.WPG

Figure 4: Position pick-up coils related to X,Y and Z gradient axis.

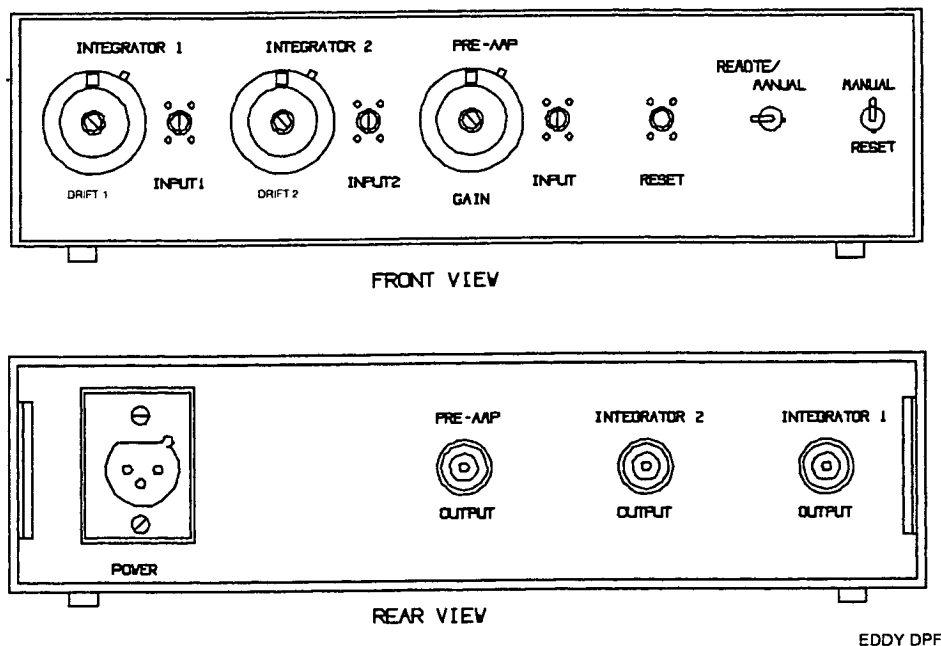
1. Mount the pick-up coils in the 15 cm positions for the 'gradient under test', see figure 3 and 4. 'Position Eddy current pick-up coils' and place the pick-up coils close to the iso centre of the bore, facing the same side.
The axis of the coils must be always parallel to the Z direction. (accuracy +/-5 degrees)
2. Do the measurement set-up, see figure 5: 'Measuring set-up for Eddy current check' and figure 6: 'Front and rear view of Philips flux meter'.
3. Switch on the fluxmeter and wait 30 minutes to stabilise.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni



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Figure 5: Measuring set-up for Eddy current check



EDDY DPF

Figure 6: Front and rear view of Philips flux meter

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

3.1. DRIFT COMPENSATION FLUX METRE

Pre-condition: Place the pick-up coils holder, with the pick-up coils in the direction, used for the gradient to be tested, on the pickup coils holder, close to the iso centre.

Procedure:

1. Set the flux meter remote/manual switch to remote.
2. The flux meter has to be switched **ON** for **at least 30 minutes**, before adjusting the offset (drift) of the flux meter.
3. Login under GYROSCAN.
 Select: <Scan Control>
 Select: <Scan Utilities>
 Select: Enter service mode <password>
 Select: System tuning <password>
 Select: System tuning tools
 Select: Installation procedures
 Select: Gradient adjustment procedures
 Select: Drift compensation flux metre.
 An information screen is displayed.
4. Select: <Proceed> To start the measurement.
5. Select: <Mod display>
 Change/check:
 monitoring mode both
 Raw data file logging no
 identifier of signal 1 Tr
 sig 1: horizontal min 1.0
 sig 1: horizontal max 4990
 sig 1: vertical min -0.1
 sig 1: vertical max 0.1
 sig 1: marker line no
 identifier of signal 2 Ti
 sig 2: horizontal min 1.0
 sig 2: horizontal max 4990
 sig 2: vertical min -0.1
 sig 2: vertical max 0.1
 sig 2: marker line no
 Select: <Proceed>
6. Select: <Display grid>
7. Select: <Display min/max>
8. Adjust, on the fluxmeter both potentiometers DRIFT 1 and 2, the drift of the both channels until the signal stays within **the limit of ± 10 mV**. (Drift is the difference between the voltage on 1 ms and 5 seconds). To see the effect of the adjustment, one must wait for at least two pulses, approx. 30s.
9. Select: <Stop scan> To stop the measurement.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni		Omni		

3.2. POSITIONING EDDY CURRENT PICK-UP COILS

Table 1

Pre-condition:

Place the pick-up coils holder, with the pick-up coils in 'xxx'-direction on the pickup coils holder, as close as possible to the iso centre.

'xxx' stands for gradient under test.

1. Select: Position eddy current pickup coils
An information screen is displayed.
2. Select: <Proceed>
3. Change/check: gradient coil Z
4. Select: <Proceed> To start the measurement.
5. If pickup coils are positioned for the X- or Y-gradient adjustment:
Select: <Mod display>
Change/check:

monitoring mode	both
Raw data file logging	no
identifier of signal 1	Tr
sig 1: horizontal min	1.0
sig 1: horizontal max	150
sig 1: vertical min	-0.1
sig 1: vertical max	0.1
sig 1: marker line	no
identifier of signal 2	Ti
sig 2: horizontal min	1.0
sig 2: horizontal max	150
sig 2: vertical min	-0.1
sig 2: vertical max	0.1
sig 2: marker line	no

Select: <Proceed>
6. If pickup coils are positioned for the X- or Y-gradient adjustment:
Move the holder with the pick-up coils in Z-direction, until the response of the two channels is:
- 50 mV < V < + 50 mV.
If pickup coils are positioned for the Z-gradient adjustment:
Move the holder with the pick-up coils in Z-direction, until the response of the two channels is:
two signals with the same amplitude (absolute).
Select: <Stop scan>
To reduce the positioning time for the other gradients, mark the position of the pickup coils holder.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni		Omni		

3.3. EDDY CURRENT CHECK

Since the Eddy currents have been adjusted in the factory, it is advised to check the actual behaviour before starting to make any readjustments. For this check, the 'eddy current final measurement' procedure will be used. During the final measurement the specific channel delay deviation is changed till **the three quarter point specification is reached**. The flatness of the averaged values are measured for the specified ranges. After completion of the final measurement the measured parameters will be checked against the specifications. A MRL will be displayed and out-of-spec values are marked with an astrix (*). If all the values are within the specification, they are stored automatically in the tuned hardware parameter archive.

You are in still within the 'Gradient adjustment procedures ' menu

1. Select: Eddy current final measurement
An information screen is displayed.
2. Select: <Proceed>
3. Change/check: gradient coil xxx
'xxx' stands for gradient under test.
4. Change/check: Current ECC settings
Activate all filters:

1	ON	2	ON	3	ON
4	ON	5	ON	6	ON
5. Select: <Proceed> To start the measurement

At the end of measurement a measurement result list (MRL) will be displayed.

Specifications for STANDARD (Copley 234_02):

ECA:	Three-quarter point	0.45 msec	71.0 % ± 2.0 %
ECA:	Flatness 1	0.90 -> 3.60 msec	< 0.15 % *)
ECA:	Flatness 2	3.60 -> 990 msec	< 0.1 % *)
ECA:	Difference 1	1.20 -> 5.60 msec	< 0.32 % **)
ECA:	Difference 2 (X or Y)	5.60 -> 990 msec (X or Y)	< 0.45 % **)
ECA:	Difference 2 (Z)	5.60 -> 990 msec (Z)	< 0.52 % **)

*) The flatness is defined as: $[(\text{max.value}) - (\text{min.value})]$

** The difference (TDIFF) is defined as: $((\text{max.value}) - (\text{min.value}))$

Specifications for OMNI (Copley 234P04):

ECA:	Three-quarter point	0.45 msec	71.0 % $\pm$ 2.0 %
ECA:	Flatness 1	0.90 -> 3.60 msec	< 0.15 % *)
ECA:	Flatness 2	3.60 -> 990 msec	< 0.1 % *)
ECA:	Difference 1	1.20 -> 5.60 msec	< 0.45 % **)
ECA:	Difference 2 (X / Y/Z)	5.60 -> 990 msec (X/ Y/Z)	< 0.42 % **)

*) The flatness is defined as: $[(\text{max.value}) - (\text{min.value})]$

The difference (TDIFF) is defined as:

$$[(\text{max.value}) - (\text{min.value})]$$

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

- When the system passes the **flatness** test (no out-of-spec values in the MRL), repeat the procedure for the next orientation. If not, continue with paragraph 4 'Eddy current adjustment procedure'
- If the system does not pass the **difference** test, it is most likely caused by the fluxmeter. To exclude this measurement tool, exchange the eddy current pickup coils connections at the front of the fluxmeter. Check the final measurement again. If difference still not within the specification contact the MR helpdesk. In the mean time change the position of the pickup coils for another axis. Repeat the pickup coils positioning and start the final measurement again.

SUMMARY of Eddy current compensation check:

X-axis

- Put the two pick-up coils to position 2 and 4 on the X-axis, see figure 3 'Position Eddy current pick-up coils'.
Do the 'Positioning eddy current pickup coils'.
Select the Z gradient and move the holder with the pick-up coils in Z-direction, until the response of the two channels is $< \pm 50$ mV.
- Do the 'Eddy current final measurement' for the X gradient.

Y-axis

- Put the two pick-up coils to position 3 and 5 on the Y-axis, see figure 3 'Position Eddy current pick-up coils'.
Do the 'Positioning eddy current pickup coils'.
Select the Z gradient and move the holder with the pick-up coils in Z-direction, until the response of the two channels is $< \pm 50$ mV.
- Do the 'Eddy current final measurement' for the Y gradient.

Z-axis

- Put the two pick-up coils to position 1 and 1' on the Z-axis, see figure 3 'Position Eddy current pick-up coils'.
Do the 'Positioning eddy current pickup coils'.
Select the Z gradient and move the holder with the pick-up coils in Z-direction, until the response of the two channels are equal, tolerance ± 100 mV.
- Do the 'Eddy current final measurement' for the Z gradient.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

4. EDDY CURRENT ADJUSTMENT PROCEDURE

4.1. INTRODUCTION

Adjustment of the magnetic field response can be done with 6 correction filters. A seventh filter A0/T0, adjusted and sealed in the factory, is not used in this procedure. The potentiometers to adjust time constant and gain factor are positioned on the WFG/ECC boards, which are located at the interface unit (DI) in the gradient amplifier cabinet. The six filters can be activated by ASW. In the Hardware parameter administration is stored which filter is active or not active. If active, the corresponding **red** led on the WFG board is on during SCAN. The filters are set during the Eddy current adjustment procedure and later on stored <STORE>.

Changing T(x) will result in another time constant and changing A(x) will lead to changing the gain factor/sign within the range of the T(x) time constant (positive or negative compensation). In the middle of the 15 turns gain potentiometer the gain is zero. By turning clock wise (CW) it gives more and more positive compensation and by turning counter clock wise (CCW) just the reverse (negative). The next table shows the different operation ranges.

Filter	Switch	Operation range (msec)	
A6/T6	Filter 6	200	3600
A5/T5	Filter 5	40	640
A4/T4	Filter 4	4	64
A3/T3	Filter 3	0.4	6.4
A2/T2	Filter 2	0.2	2.2
A1/T1	Filter 1	0.03	0.33

A filter setting (A,T) must be adjusted in case shape A is found, i.e. an overshoot followed by an undershoot. If this shape is found, both potentiometers must be adjusted iterative T(x) (CW) and A counter (CCW) to reduce the overshoot and undershoot. But be aware: never loose the principle shape overshoot followed by a undershoot.

Start with turning the gain potentiometer (A) of the filter to be adjusted clockwise till a clear overshoot can be seen.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni		Omni		

Depending on the shape of the overshoot a certain correction procedure must be chosen.
 In part A of figure 7, the time constant is too small, increase T time constant potentiometer.
 In part B of figure 7, the time constant is too large and/or the amplitude is too high. Change both potentiometers to correct it

FIGURE A FIELD RESPONSE TIME
CONSTANT TOO HIGH

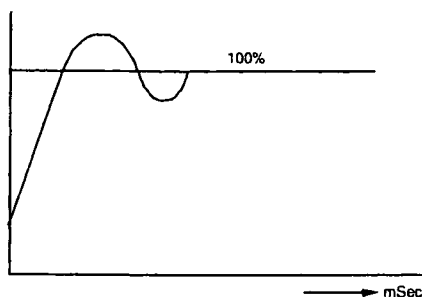
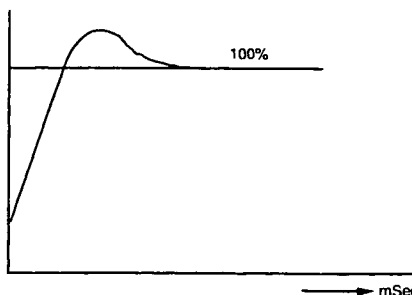


FIGURE B FIELD RESPONSE TIME
CONSTANT TOO LARGE
AND/OR AMPLITUDE TOO HIGH



HH11 WPG

Figure 7: Field response as function of time.

Pre-condition:

- The pick-up coils are placed in direction under test in the marked position of the holder. Before starting the Eddy current correction procedure it is advisable to do a **'REFERENCE MEASUREMENT'**. During this adjustment the gradient channel delay parameters, determined in the procedures 'Eddy current final measurement' and/or 'Gradient channel delay determination', will be set automatically to the default value 0.025 msec.

Start-up procedure for the following paragraphs.

- Login under GYROSCAN.
 - Select: <Scan Control>
 - Select: <Scan Utilities>
 - Select: Enter service mode <password>
 - Select: System tuning <password>
 - Select: System tuning tools
 - Select: Installation procedures
 - Select: Gradient adjustment procedures

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
Standard	Omni	Omni

4.2. SETTING STARTING POINT FOR THE POTENTIOMETERS (OPTIONAL)

After installing a new board or in case the flatness of the eddy-current compensation is completely out of specification, it is advised to return the potentiometers to the starting position (= centre =zero position). In general however, it should be sufficient to start the adjustment without turning the potentiometers first.

Turning the potentiometers to the starting position:

Turn the 6 A's and 6 T's potentiometers of X,Y or Z WFG/ECC board fully CCW and after that turn all A's 7.5 turns CW. (gain zero). An overshoot can be expected in this starting condition.

4.3. EDDY CURRENT REFERENCE MEASUREMENT

Pre-condition: The pick-up coils are placed in the direction under test.

Procedure:

1. Select: Eddy current reference measurement
An information screen is displayed.
2. Select: <Proceed>
3. Change/check:

Measuring type	Reference	
Gradient coil	Gradient under test	equals position of the pick-up coils
Gradient ampl. faults	enabled	
4. Select: <Proceed> to start the measurement.
The reference measurement is performed and the maximum gradient strength is measured.
The program will display the result.

The maximum gradient strength should be: **> 4.5 mT/m**

5. Enter: <RETURN> to accept this condition.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni		Omni		

In case an overshoot is followed by an undershoot, then:

- Turn T6 (+TIME) CW till the undershoot is reduced till 1/16 of the overshoot and turn A6 (-GAIN) CCW till the amplitude of the overshoot above the 100% level is equal to the amplitude of the undershoot under the 100% level. Repeat this action till the optimum result is reached.
- The best Eddy current compensation is reached by maximum gain (A) and minimum time (T) settings.
- If necessary modify the vertical scaling factor, select with mouse 'Mod Display'. This also applies to the other filter combinations.

7. Select: <Stop scan>

8. Change/check: Current ECC settings

1	OFF	2	OFF	3	OFF
4	OFF	5	ON	6	ON

9. Select: <Proceed> twice To start the adjustment of filter 5.

10. Adjustment of the A5/T5 filter. Check the flatness. If the flatness is not correct try to improve it by tuning only the potentiometer A5 a little bit. Do not yet touch T5.

The flatness should be < 0.1 %, where the flatness is defined as: $[(\text{max.value}) - (\text{min.value})]$

When you do not succeed with this adjustment then turn T5 a little bit clock wise and turn A5 clock wise (+GAIN) till the last part of the displayed curve is flat as possible, but at least within the specifications.

While turning A5 CW you are only able to lift-up the curve. If already a overshoot exists use the negative compensation turning A5 CCW (-GAIN). If within specifications, continue with the next point.

11. Select: <Stop scan>

12. Change/check: Current ECC settings

1	OFF	2	OFF	3	OFF
4	ON	5	ON	6	ON

13. Select: <Proceed> twice To start the adjustment of filter 4.

14. Adjustment of the A4/T4 filter. Check the flatness. If the flatness is not correct try to improve it by tuning only the potentiometer A4 a little bit. Do not yet touch T4.

The flatness should be < 0.1 %, where the flatness is defined as: $[(\text{max.value}) - (\text{min.value})]$

When you do not succeed with this adjustment then turn T4 a little bit clock wise and turn A4 clock wise (+GAIN) till the last part of the displayed curve is flat as possible, but at least within the specifications.

While turning A4 CW you are only able to lift-up the curve. If already a overshoot exists use the negative compensation turning A4 CCW (-GAIN). If within specifications, continue with the next point.

15. Select: <Stop scan>

16. Select: <Cancel>

17. Select: Compensate mid term eddy current
An information screen is displayed.

18. Select: <Proceed>

19. Change/check: Current ECC settings

1	OFF	2	OFF	3	ON
4	ON	5	ON	6	ON

20. Select: <Proceed> twice To start the adjustment of filter 3.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni		Omni		

21. Adjustment of the A3/T3 filter. Check the flatness. If the flatness is not correct try to improve it by turning only the potentiometer A3 a little bit. Do not yet touch T3.

The flatness should be (where the flatness is defined as: $[(\text{max.value}) - (\text{min.value})]$):

From 1 ms until 3.6 ms: **< 0.15 %**

From 3.6 ms onwards: **< 0.1 %**

When you do not succeed with this adjustment then turn T3 a little bit clock wise and turn A3 clock wise (+GAIN) till the last part of the displayed curve is flat as possible, but at least within the specifications.

While turning A3 CW you are only able to lift-up the curve. If already a overshoot exists use the negative compensation turning A3 CCW (-GAIN). If within specifications, continue with the next point

22. Select: <Stop scan>
 23. Select: <Cancel>
 24. Select: Compensate short term eddy current
 25. Change/check: Current ECC settings

1	OFF	2	ON	3	ON
4	ON	5	ON	6	ON

26. Select: <Proceed> twice To start the adjustment of filter 2.

27. Select: <Mod display>

Change/check:

monitoring mode	one
raw data logging	no
identifier of signal 1	Tavg
sig 1: horizontal min	0.90
sig 1: horizontal max	5.00
sig 1: vertical min	99.8
sig 1: vertical max	100.2
sig 1: marker line	no

Select: <Proceed>

28. Adjustment of the A2/T2 filter. Check the flatness. If the flatness is not correct try to improve it by turning only the potentiometer A2 a little bit. Do not yet touch T2.

The flatness should be (where the flatness is defined as: $[(\text{max.value}) - (\text{min.value})]$):

From 0.9 ms until 3.6 ms: **< 0.15 %**

From 3.6 ms onwards: **< 0.1 %**

Check also the response at 0.45 msec, should be: 71.0 % $\pm$ 2.0 %. The response is displayed on the 'menu' selection window. Response value is coloured RED if not within specifications.

When you do not succeed with this adjustment then turn T2 a little bit clock wise and turn A2 clock wise (+GAIN) till the displayed curve is flat as possible, but at least within the specifications. While turning A2 CW you are only able to lift-up the curve. If already a overshoot exists use the negative compensation turning A2 CCW (-GAIN). If within specifications, continue with the next point.

29. Select: <Stop scan>
 30. Change/check: Current ECC settings

1	ON	2	ON	3	ON
4	ON	5	ON	6	ON

31. Select: <Proceed> twice To start the adjustment of filter 1.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni		Omni		

32. Select: <Mod display>

Change/check:

monitoring mode one
raw data logging no
identifier of signal 1 Tavg
sig 1: horizontal min 0.90
sig 1: horizontal max 5.00
sig 1: vertical min 99.8
sig 1: vertical max 100.2
sig 1: marker line no

Select: <Proceed>

33. Adjust by means of an iterative process A1 and T1, in such a way that the rise time and flatness of the curve meets the specifications.

The flatness should be (where the flatness is defined as: $[(\text{max.value}) - (\text{min.value})]$):

From 0.9 ms until 3.6 ms: < 0.15 %

From 3.6 ms onwards: < 0.1 %

Check also the response at 0.45 msec, should be: 71.0 % $\pm$ 2.0 %. The response is displayed on the 'menu' selection window. Response value is coloured RED if not within specifications.

If signal is too high (small): turn A1 CCW (CW) to reach the rise time specification and T1 CW (CCW) to improve the flatness of the curve.

34. Select: <Stop scan>

35. Select: <Cancel>

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni		Omni		

4.5. EDDY CURRENT FINAL MEASUREMENT

1. Select: Eddy current final measurement
An information screen is displayed.
 2. Select: <Proceed>
 3. Change/check: gradient coil xxx
 'xxx' stands for gradient under test.
 4. Change/check Settings
- | | | | | | |
|---|----|---|----|---|----|
| 1 | ON | 2 | ON | 3 | ON |
| 4 | ON | 5 | ON | 6 | ON |
5. Select: <Proceed> To start the measurement.

At the end of measurement a measurement result list (MRL) will be displayed. For specifications see paragraph 3.3.

- If the system does not pass the **flatness** tests, check the times of the extreme values. Depending on these times a part of paragraph 4 'Eddy current adjustment procedure' must be repeated. Otherwise change the position of the pickup coils for another axis. Repeat the pickup coils positioning and start the final measurement again.
- If the system does not pass the **difference** tests, then the most likely cause is the fluxmeter. To exclude the measurement tool, exchange the eddy current probe connections at the front of the fluxmeter. Check the final measurement again. If difference still not within the specification contact the MR helpdesk. In the mean time change the position of the pickup coils for another axis. Repeat the pickup coils positioning and start the final measurement again.

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni		Omni		

4.6. EXAMPLE OF MRL PRINT OUT

Example, with some comments added.

Line	Description		Final	Nominal	Spec
1.	measurement type		Final		Spec
2.	gradient coil X,Y or Z		Y		
3.	current ECC settings Y	1	<value>		
4.	current ECC settings Y	2	<value>		
5.	current ECC settings Y	3	<value>		
6.	current ECC settings Y	4	<value>		
7.	current ECC settings Y	5	<value>		
8.	current ECC settings Y	6	<value>		
9.	settings Y	1	<value>		
10.	settings Y	2	<value>		
11.	settings Y	3	<value>		
12.	settings Y	4	<value>		
13.	settings Y	5	<value>		
14.	settings Y	6	<value>		
15.	overall gradient delay (msec)		<value>		
16.	delay deviation (us)		<value>	0.00	[-12.0 - 12.0]
17.	three-quarter point (0.45 msec)		<value>	71.00	[69.00 - 73.00]
19.	Flatness range 1 (%)		<value>	0.00	< 0.15
20.	Flatness range 2 (%)		<value>	0.00	< 0.10
21.	Difference 1 (%)		<value>	0.00	< 0.32 (STANDARD), < 0.45 (OMNI)
22.	Difference 2 (%) X or Y		<value>	0.00	< 0.45 (STANDARD); < 0.42 (OMNI)
	Difference 2 (%) Z		<value>	0.00	< 0.52 (STANDARD); < 0.42 (OMNI)
23.	Extreme times (ms) 1		<value>		
24.	Extreme times (ms) 2		<value>		
25.	Extreme times (ms) 3		<value>		
26.	Extreme times (ms) 4		<value>		
27.	Extreme times (ms) 5		<value>		
28.	Extreme times (ms) 6		<value>		
29.	Extreme times (ms) 7		<value>		
30.	Extreme times (ms) 8		<value>		
31.	Spec. ranges (ms) 1		0.30		
32.	Spec. ranges (ms) 2		3.00		
33.	Spec. ranges (ms) 3		3.00		
34.	Spec. ranges (ms) 4		990.00		
35.	Spec. ranges (ms) 5		0.60		
36.	Spec. ranges (ms) 6		5.00		
37.	Spec. ranges (ms) 7		5.00		
38.	Spec. ranges (ms) 8		990.00		
39.	Meas-system delay [us]		24.00	24.00	[23.50 - 24.50]

Intera 0.5 T		Intera 1.0 T		Intera 1.5 T	
Standard	Omni		Omni		

Remarks:

In the actual display of the result screen the line numbers are not shown.

Line

Line 23	Time at which minimum of averaged signal is measured for flatness 1
Line 24	Time at which maximum of averaged signal is measured for flatness 1
Line 25	Time at which minimum of averaged signal is measured for flatness 2
Line 26	Time at which maximum of averaged signal is measured for flatness 2
Line 27	Time at which minimum of difference signal is measured for difference 1
Line 28	Time at which maximum of difference signal is measured for difference 1
Line 29	Time at which minimum of difference signal is measured for difference 2
Line 30	Time at which maximum of difference signal is measured for difference 2
Line 31	Start time specification range for flatness 1.
	STANDARD/OMNI: add 0.6 ms to find actual start time
Line 32	End time specification range for flatness 1.
	STANDARD/OMNI: add 0.6 ms to find actual end time
Line 33	Start time specification range for flatness 2.
	STANDARD/OMNI: add 0.6 ms to find actual start time
Line 34	End time specification range for flatness 2.
Line 35	Start time specification range for difference 1
	STANDARD/OMNI: add 0.6 ms to find actual start time
Line 36	End time specification range for difference 1
	STANDARD/OMNI: add 0.6 ms to find actual end time
Line 37	Start time specification range for difference 2
	STANDARD/OMNI: add 0.6 ms to find actual start time
Line 38	End time specification range for flatness difference 2
Line 39	Meas-system delay

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Section 17

Gradient system

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	Power	Power Master

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Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

1. SETTING TO WORK THE GRADIENT SYSTEM

WARNING

Be aware of the high voltage (420 Volt) present inside the gradient amplifier cabinet(s), on the connections to the gradient coil and the system filter box. Make sure that the AC power has been switched off for at least 2 minutes before starting to work on the gradient chain. It is advised to watch the high voltage value on the EMI 20kW/420V DC power supply meters and make sure that the supply capacitors have discharge to less than 2 Volt.

1.1. CONFIGURATION GRADIENT AMPLIFIERS CABINETS

POWER:

The gradient system POWER consist of one cabinet. This cabinet contains the gradient interface rack (DI) with the gradient interface controller and the 3 waveform generator WFG/ECC 2 boards, followed by 3 Copley 274 amplifier axes, a power distribution unit (PDU) and two parallel switched EMI 20 kW/420 V DC power supplies (a master and a slave unit).

MASTER:

The gradient system MASTER consists of two cabinets - a master and a slave. The master cabinet contains the gradient interface rack (DI) with the gradient interface controller and the 3 waveform generator WFG/ECC 2 boards, followed by 3 Copley 274 amplifier axes, a power distribution unit (PDU) and two parallel switched EMI 20 kW/420 V DC power supplies (again a master and a slave). The slave gradient cabinet has the same configuration as the master cabinet except for the gradient interface rack.

Configuration gradient coils:

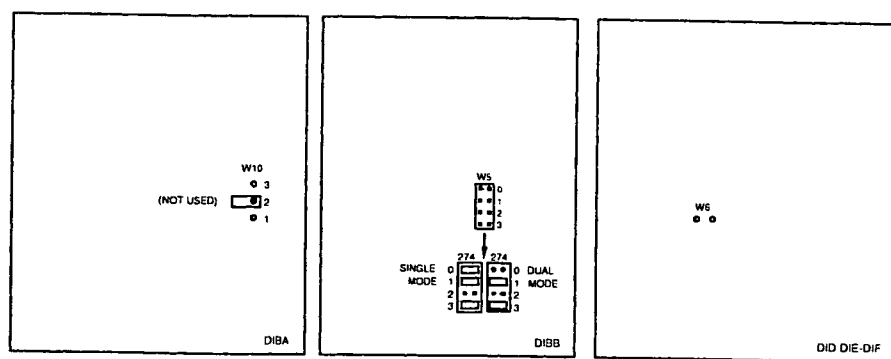
POWER/MASTER Water cooled gradient coil.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

1.2. PRE POWER-UP CHECKS PROGRAMMING

Before applying power to the cabinet(s) make sure that:

1. the jumper-setting of the power modules and the controller is according to tables 1,2,3a,3b and 3c. (jumper-setting can be seen from the back side of the cabinets.)
2. the jumper-setting of the Gradient Interface Controller (DIBA/DIBB) for single (POWER) or dual mode (MASTER) is according figure 1.
3. for POWER/MASTER, the jumper W6 of the three WFG/ECC2 boards (DID/DIE/DIF) is not installed see figure 1.



HH81 WPG

Figure 1: Jumpers setting of the Gradient Interface boards.

4. the gradient system inter connections are tightened properly.
5. the grounding straps of the side panels and doors of the cabinets have been checked.
6. all power module fan assemblies are properly mounted.
7. the back doors are closed.

CAUTION

*Before switching on the cabinets, all magnet covers must be mounted, the water cooling must be operational, and the temperature sensor on the gradient-coil must be connected.
If covers are not yet mounted or the RF room is not operational, overrule temporally the gradient air-pressure sensor. Make a connection between TFEI connector MDI5X1: pin # 1 to pin # 6.*

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

 PUSHED TO FRONT TO LEFT = "1"

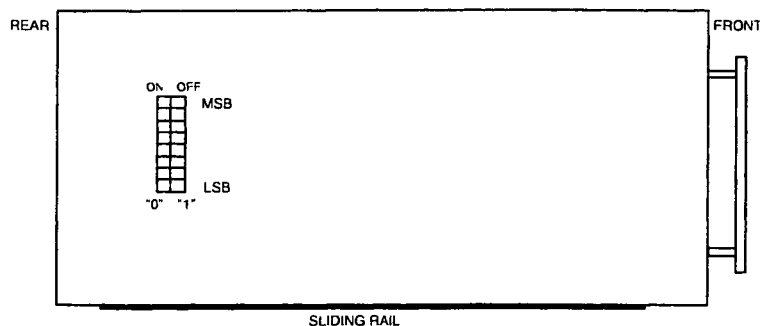


Figure 2: Dip switches location Copley-274 Power Module

Dip switches located on the Copley-274 Power Module		
FUNCTION	Master cabinet	Dip switches (MSB-LSB)
address	power module 1	00000001
address	power module 2	00000010
address	power module 3	00000011
address	power module 4	00000100
address	power module 5	00000101
address	power module 6	00000110
address	power module 7	00000111
address	power module 8	00001000
address	power module 9	00001001
address	power module 10	00001010
address	power module 11	00001011
address	power module 12	00001100
Address dip switches located on the Copley-274 Controller * see remark		
address	controller X 13	dddduudu
address	controller Y 14	dddduuud
address	controller Z 15	dddduuuu

Table 1: Address selection of the the master cabinet.

Remark: In some controllers the dip switches of the address selection are mirrored (MSB - LSB), see figure 4. Always check the label, behind the 'door' of the controller panel.

Intera 0 5 T	Intera 1 0 T	Intera 1 5 T
	Power	Power Master

Dip switches located on the Copley-274 Power Module		
FUNCTION	Slave cabinet	Dip switches (MSB-LSB)
address	power module 17	00010001
address	power module 18	00010010
address	power module 19	00010011
address	power module 20	00010100
address	power module 21	00010101
address	power module 22	00010110
address	power module 23	00010111
address	power module 24	00011000
address	power module 25	00011001
address	power module 26	00011010
address	power module 27	00011011
address	power module 28	00011100
Address dip switches located on the Copley-274 Controller * see remark		
address	controller X 29	ddduuuu
address	controller Y 30	ddduuuud
address	controller Z 31	ddduuuuu

Table 2: Address selection of the slave cabinet.

Remark: In some controllers the dip switches of the address selection are mirrored (MSB - LSB), see figure 4. Always check the label, behind the 'door' of the controller panel.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Normal, in most controllers, lay-out of the dip switches of the address selection:

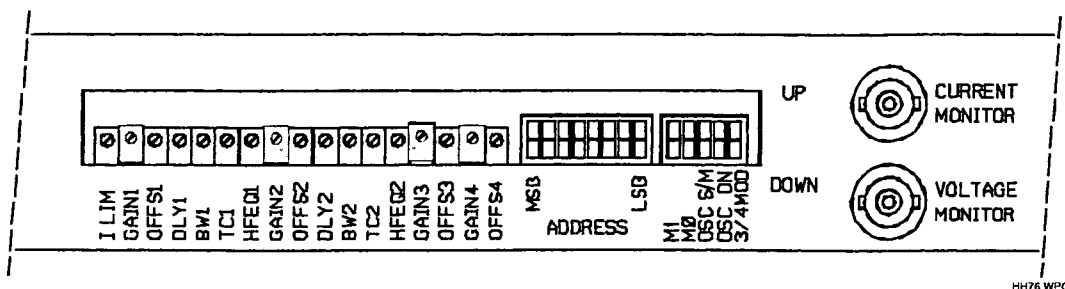


Figure 3: Dip switch location Copley-274 Controller

Remark: In some controllers the dip switches of the address selection are mirrored (MSB - LSB), see figure 4. Always check the label, behind the 'door' of the controller panel. Also on these panels the M0 and M1 switches are mirrored.

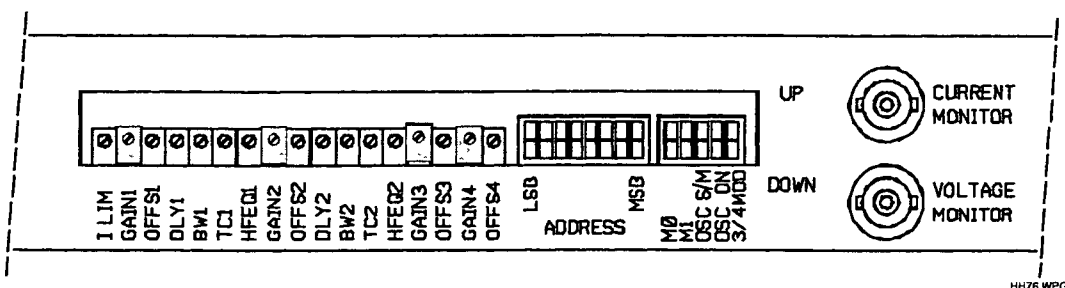


Figure 4: Dip switch location Copley-274 Controller (mirrored)

FUNCTIONS	POWER	Copley-274 Controller dip switches M1,M0,OSC S/M,OSC ON,3/4 MOD
coil mode(s), clock	controller X,Y and Z	dduud

Table 3a: Coil_1 mode- and clock signal selection of the POWER cabinet.

FUNCTIONS	MASTER Master and Slave	Copley-274 Controller dip switches M1,M0,OSC S/M,OSC ON,3/4 MOD
coil mode(s), clock	controller X,Y and Z	duuud ' see remark

Table 3b: Coil_2 mode- and clock signal selection of the PT 6000 master- and slave cabinets.

Remark: In some controllers the dip switches M0 and M1 are mirrored, see figure 4. Always check the label, behind the 'door' of the controller panel.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

1.3. PRE POWER-UP CHECKS GRADIENT CABLE CONNECTIONS

1.3.1. PROCEDURE POWER

Before switching ON the gradient amplifier, one must check the gradient cable connections for any shorts. Checks are done with a digital voltmeter (DVM):

- the resistance of gradient coil must be measured;
- the resistance between the gradient coils must be measured;
- the resistance from the cable connections to ground must be measured.

WARNING

For this measurement the gradient amplifier must be switched OFF. In case the gradient amplifier is switched ON, then switched it OFF. Make sure that the AC power has been switched off for at least 2 minutes before starting to work on the gradient chain. It is advised to watch the high voltage value on the EMI 20kW/420V DC power supply meters and make sure that the supply capacitors have discharge to less than 2 Volt.

CABLES NOT DRAWN BUT MUST HAVE BEEN CONNECTED

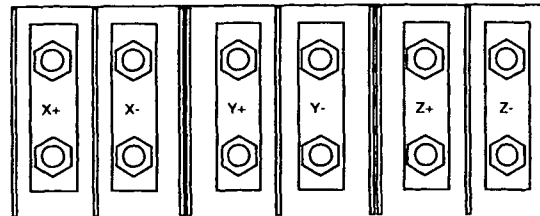


Figure 5: Gradient cable connections at rear of the gradient amplifier

In table 4, the measurement points are given. The type of measurement refers to:

- the resistance of gradient coil;
- the resistance between the gradient coils;
- the resistance from the cable connections to ground.

Type	DVM (Ohm range) between:		Spec. [Ω]	Measured [Ω]
a1	X+	X-	< 10	
a2	Y+	Y-	< 10	
a3	Z+	Z-	< 10	
b1	X+	Y+	> 10	
b2	X+	Z+	> 10	
b3	Y+	Z+	> 10	
c1	X+	ground	> 10	
c2	Y+	ground	> 10	
c3	Z+	ground	> 10	

Table 4: POWER gradient cable connections checks.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

1.3.2. PROCEDURE MASTER

Before switching ON the gradient amplifier, one must check the gradient cable connections for any shorts. Checks are done with a digital voltmeter (DVM):

- the resistance of gradient coil must be measured;
- the resistance between the gradient coils must be measured,
- the resistance from the cable connections to ground must be measured.

WARNING

For this measurement the gradient amplifier must be switched OFF. In case the gradient amplifier is switched ON, then switched it OFF. Make sure that the AC power has been switched off for at least 2 minutes before starting to work on the gradient chain. It is advised to watch the high voltage value on the EMI 20kW/420V DC power supply meters and make sure that the supply capacitors have discharge to less than 2 Volt.

CABLES NOT DRAWN BUT MUST HAVE BEEN CONNECTED

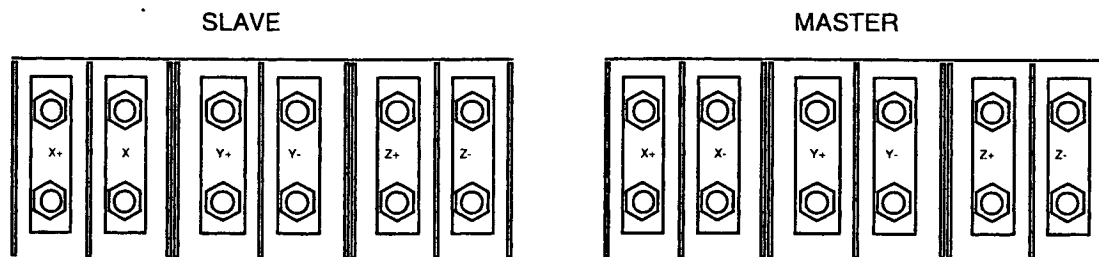


Figure 6: Gradient cables connections at rear of the gradient amplifier cabinets

In table 5, the measurement points are given. The type of measurement refers to:

- the resistance of gradient coil;
- the resistance between the gradient coils;
- the resistance from the cable connections to ground.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Type	DVM (Ohm range) between:		Spec.[Ω]	Measured[Ω]
a1	X+ master	X- master	< 10	
a2	Y+ master	Y- master	< 10	
a3	Z+ master	Z- master	< 10	
a4	X+ slave	X- slave	< 10	
a5	Y+ slave	Y- slave	< 10	
a6	Z+ slave	Z- slave	< 10	
b1	X+ master	Y+ master	> 10	
b2	X+ master	Z+ master	> 10	
b3	Y+ master	Z+ master	> 10	
b4	X+ slave	Y+ slave	> 10	
b5	X+ slave	Z+ slave	> 10	
b6	Y+ slave	Z+ slave	> 10	
b7	X+ master	Y+ slave	> 10	
b8	X+ master	Z+ slave	> 10	
b9	Y+ master	X+ slave	> 10	
b10	Y+ master	Z+ slave	> 10	
b11	Z+ master	X+ slave	> 10	
b12	Z+ master	Y+ slave	> 10	
b13	X+ master	X+ slave	> 10	
b14	Y+ master	Y+ slave	> 10	
b15	Z+ master	Z+ slave	> 10	
c1	X+ master	ground	> 10	
c2	Y+ master	ground	> 10	
c3	Z+ master	ground	> 10	
c4	X+ slave	ground	> 10	
c5	Y+ slave	ground	> 10	
c6	Z+ slave	ground	> 10	

Table 5: MASTER gradient cable connections checks.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

1.4. STT/TSW SETTINGS

Verify Gradient amplifier parameters/drivers in ASW

Login under GYROTOOL

Select: Modify system configuration

Select: Scanner Hardware configuration

Select: Grad. amplifier type

Enter: For POWER Copley_274S

For MASTER Copley_274D

Select: Grad. coil type

Enter: For POWER/MASTER Watercooled

Select: <Proceed>

Exit from GYROTOOL

Login under GYRORESTART

Restart all processes by selecting to re-start the logging process. Enter: yes.

Verify Gradient amplifier drivers in TSW

Login under TSW

Select: CONF

Select: UPDATE

Enter: <operators name> name of the operator

Enter: <system> name hospital

Enter: Y update the configuration file with site-specific information

The TSW configuration file is modified with the configuration data as set in the application software.

Enter: <RETURN> After update ready message.

Select: INSPECT

Verify: Line 696. Which gradient amplifier type is set in the configuration.

The value must be 'copley-274D for MASTER or 'copley-274S for POWER.

Select: QUIT

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

1.5. FIRST TIME POWER-ON CHECKS

1. Make sure that NTDAC has been switched on (otherwise the sync_clock is missing).
2. Switch on gradient amplifier in the MDU (MDE).
3. Switch on the MAINS and AUX switches of the master and slave (MASTER only) gradient amplifier cabinet. The switches can be found on both the master and slave PDU's.
4. All power module fan units should run.
5. Switch on both (for MASTER = 4) EMI 20 kW/420V power supplies.
6. The heat exchanger for the gradient coil must be on. (POWER/MASTER only)
7. Verify the status of the heat exchanger: (POWER/MASTER only)

Temp. control dial: 25 ° C
 Recirculation temperature: equal to the dial setting (tolerance)
 Recirculation pressure: 43.5 PSI - 79.8 PSI (3 - 5 bar)
 Pressure fluctuation: 7.2 PSI (0.5 bar)
 Water purity light: After (re)filling the secondary circuit, the light will shine red for a couple of days. This indicates that the ion-level is not yet within specification. The light should change to green within one week. Note that during this first week, it is allowed to operate the system with the red-led lit. When after normal operation, the lamp changes to steady red the cartridge should be replaced. If the light is periodically changing red/green, the purity is still adequate but replacement should be anticipated.

Idle- & cool light: The amber IDLE and the green COOL lights indicate the status of the control valve. When the temperature control valve is fully open (for maximum cooling), the COOL light is on steady. When the control valve is fully closed the IDLE light is on. As the control valve moves between these extremes, the two lights flash with variable on-times to indicate the approximate position of the control valve.

8. Login under GYROSCAN.
 Select: <Scan Control>
 Select: <Scan Utilities>
 Select: Enter service mode <password>
 Select: System tuning <password>
 Select: System tuning tools
 Select: Installation Procedures
 Select: Gradient chain (274) measurements

Status check:

Procedure POWER:

1. Select: Gradient status check
 An information screen is displayed.
2. Select: <Proceed> To start the measurement.
 At the end of the test a MRL is displayed. The EMI power check must return 'success' status.
3. Select: <Proceed>

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Procedure MASTER:

1. Select: Master tests. Status check
An information screen is displayed.
2. Select: <Proceed> To start the measurement.
At the end of the test a MRL is displayed. The EMI power check must return 'success' status.
3. Select: <Proceed>
4. Select: Slave tests. Status check
An information screen is displayed
5. Select: <Proceed> To start the measurement.
At the end of the test a MRL is displayed. The EMI power check must return 'success' status.
6. Select: <Proceed>

Analog function test:**Procedure POWER/MASTER:**

1. Select: Analog functions test
An information screen is displayed.
2. Select: <Proceed> To start the measurement.
At the end of the test a MRL is displayed. No errors are allowed.
3. Select: <Proceed>

Gradient load test:**Procedure POWER:**

1. Select: Gradient load test
An information screen is displayed.
2. Select: <Proceed> To start the measurement.
At the end of the test a MRL is displayed. No errors are allowed.
3. Select: <Proceed>
4. Select: <Cancel> twice

Procedure MASTER:

1. Select: Gradient load test (master)
An information screen is displayed.
2. Select: <Proceed> To start the measurement.
At the end of the test a MRL is displayed. No errors are allowed.
3. Select: <Proceed>
4. Select: Gradient load test (slave)
An information screen is displayed.
5. Select: <Proceed>
At the end of the test a MRL is displayed. No errors are allowed.
6. Select: <Proceed>
7. Select: <Cancel> twice

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

2. MEASUREMENTS ON THE GRADIENT SYSTEM

2.1. OFFSET CHECK

This test measures the offset values of the Copley-274 Amplifiers in enabled mode.

Procedure:

1. Login under GYROSCAN.
 - Select: <Scan Control>
 - Select: <Scan Utilities>
 - Select: Enter service mode <password>
 - Select: System tuning <password>
 - Select: System tuning tools
 - Select: Installation Procedures
 - Select: Gradient chain (274) measurements

Procedure POWER:

2. Select: Gradient controller gain/offset test
An information screen is displayed.
3. Select: <Proceed> To start the measurement
The test is done. At the end of the test a MRL is displayed. Check that all values are within the limits.
4. Select: <Proceed>
5. Select: <Cancel> twice

Procedure MASTER:

2. Select: Master tests: Controller gain/offset
An information screen is displayed.
3. Select: <Proceed> To start the measurement.
The test is done. At the end of the test a MRL is displayed. Check that all values are within the limits.
4. Select: <Proceed>
5. Select: Slave tests: Controller gain/offset
6. Select: <Proceed> To start the measurement
The test is done. At the end of the test a MRL is displayed. Check that all values are within the limits.
7. Select: <Proceed>
8. Select: <Cancel> twice

If test results is not within the specifications, then one need to trouble-shoot the gradient amplifier.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

2.2. POWER GRADIENT CURRENT ADJUSTMENT (COPLEY GAIN ADJUSTMENT)

The 'Gradient current adjustment' procedure is used to check/adjust the gain setting of the Copley-274 Gradient Amplifiers. The test uses a demand signal that should produce a coil current of 240 A with a tolerance of ± 3.6 A in each channel. The three axis currents for a POWER are measured, and the results are displayed on the console monitor.

Precondition: The RF door should be closed.

Procedure.

1. Login under GYROSCAN.
 Select: <Scan Control>
 Select: <Scan Utilities>
 Select: Enter service mode <password>
 Select: System tuning <password>
 Select: System tuning tools
 Select: Installation Procedures
 Select: Gradient adjustment procedures
 Select: Gradient current adjustment.
 An information screen is displayed
2. Select: <Proceed> To start the measurement.

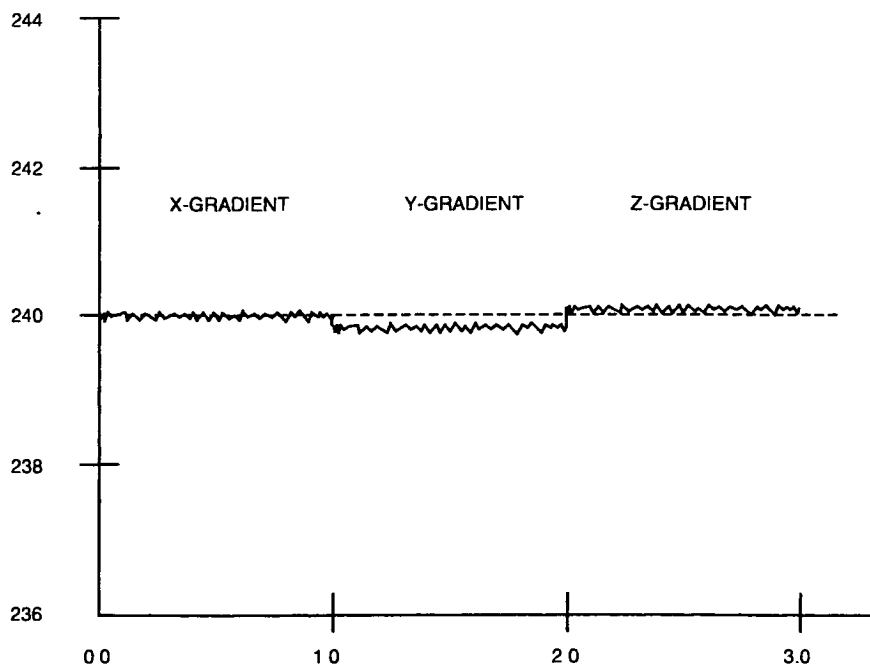


Figure 7: Gradient current adjustment, displayed on the console monitor

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

The adjustment of the currents is done with 'GAIN1' potentiometer of the corresponding Copley-274 Controller.

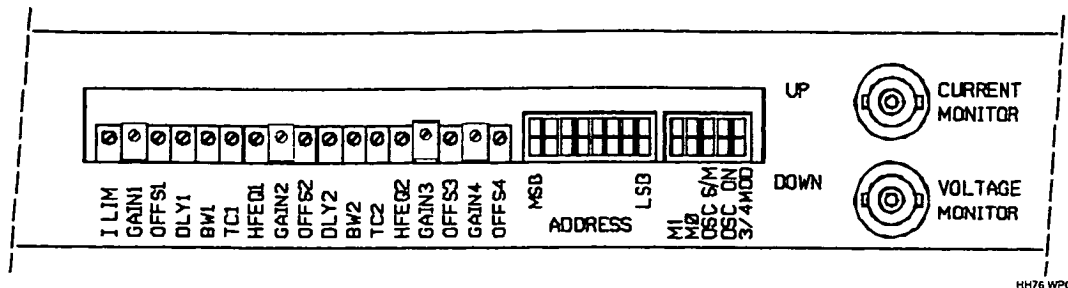


Figure 8: 'GAIN1' potentiometer location Copley-274 Controller

When adjustments are within the limits, preferable as close as possible to the 240 A value:

3. Enter: Exit
4. Select: <Cancel> Twice To exit the 'gradient adjustment procedures' menu.

2.3. MASTER GRADIENT CURRENT ADJUSTMENT (COPLEY GAIN ADJUSTMENT)

The 'Gradient current adjustment' procedure is used to check/adjust the gain setting of the Copley-274 Gradient Amplifiers. The test uses a demand signal that should produce a coil current of 240 A with a tolerance of ± 3.6 A in each channel. For a MASTER the six axis currents are measured, and the results are displayed on the console monitor.

Precondition: The RF door should be closed.

Procedure:

1. Login under GYROSCAN.
 - Select: <Scan Control>
 - Select: <Scan Utilities>
 - Select: Enter service mode <password>
 - Select: System tuning <password>
 - Select: System tuning tools
 - Select: Installation Procedures
 - Select: Gradient adjustment procedures
 - Select: Gradient current adjustment.

An information screen is displayed. Follow the screen messages.
2. Select: <Proceed> To start the measurement.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

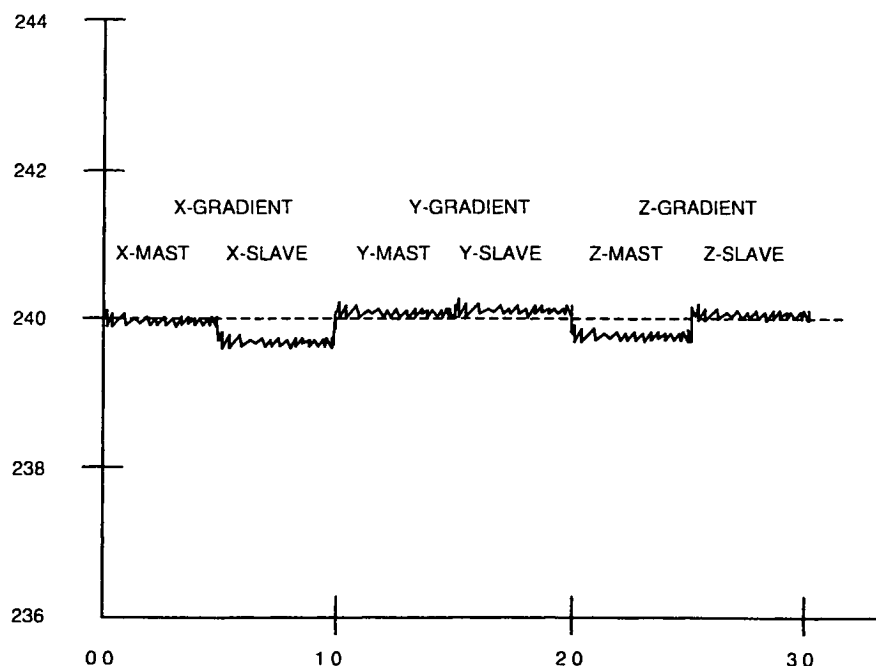


Figure 10: MASTER gradient current adjustment, displayed on the console monitor

The adjustment of the currents is done with 'GAIN2' potentiometer of the corresponding Copley-274 Controller.

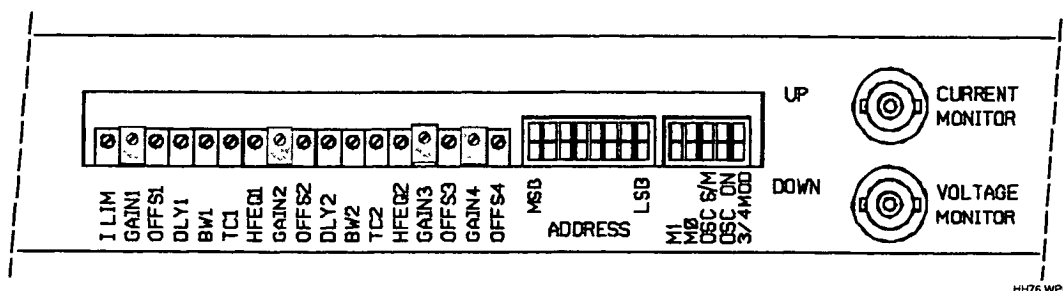


Figure 11: 'GAIN2' potentiometer location Copley-274 Controller

When adjustments are within the limits, preferable as close as possible to the 240 A value:

3. Enter: Exit
4. Select: <Cancel> Twice To exit the 'gradient adjustment procedures' menu.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

2.4. SETTLING, INTRODUCTION

The 'settling' is the behaviour of the actual gradient pulse, shortly after a slope. With respect to the demand pulse the actual gradient will trail and tend to oscillate (damped oscillation). In order to obtain a defined response of the gradient pulse the settling adjustment procedure is implemented. An incorrect adjusted settling behaviour will cause difficulties with the 'short-time' eddy current compensation adjustment and can cause gradient amplifier shutdowns.

Principle of adjustment

The 'settling' adjustment procedure uses three pulses:

- demand pulse, this is the pulse sent to the Copley-274 amplifier;
 - reference gradient response, this is a by the computer calculated pulse. The wanted response from the gradient on a demand pulse is calculated. In order to calculate this reference a difference with respect to the demand pulse is used. The differences are:
 - at 100 usec after the end of slope: 99.5 %
 - at 150 usec after the end of slope: 99.8 %
 - at 250 usec after the end of slope: 99.9 %
- The demand pulse is considered to be 100% after the end of slope.
- the actual gradient response, which is scaled to 100%.

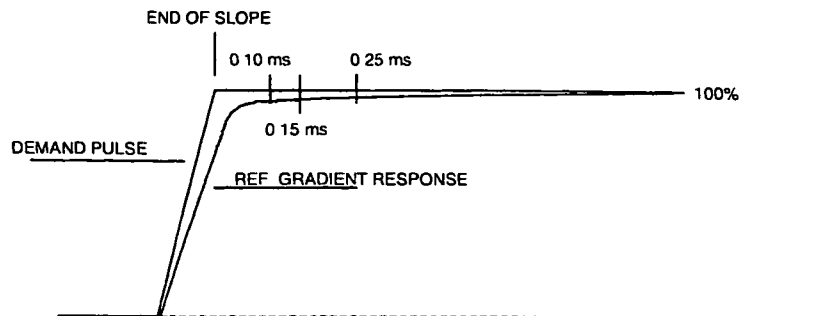


Figure 12: Demand pulse and reference gradient pulse.

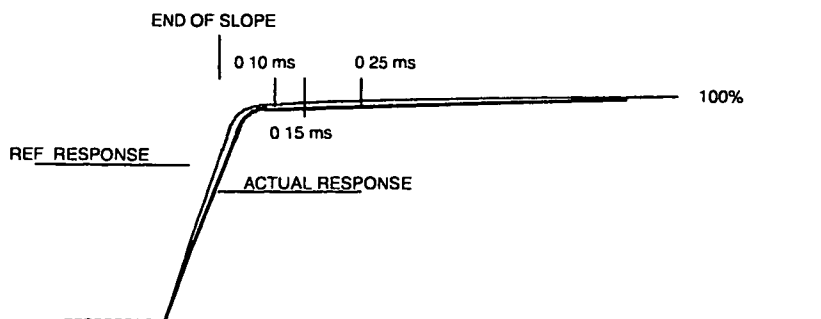


Figure 13: Reference gradient response and actual gradient pulse.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

The purpose is to adjust the actual gradient response as close as possible to the reference gradient response. During the adjustment the difference (difference between reference and actual response) signal is displayed and used for the settling adjustment. The adjustment is done with the potentiometers 'TC1', 'BW1' and 'HFEQ1' for POWER, and with the potentiometers 'TC2', 'BW2' and 'HFEQ2' for MASTER. These potentiometers are located on each Copley-274 Controller.

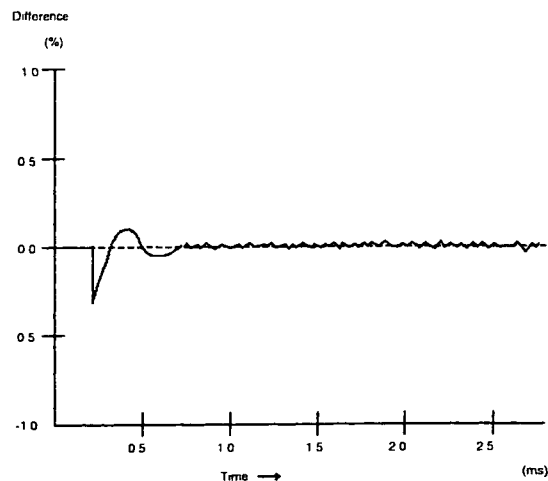


Figure 14: Difference between actual and reference gradient response.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Explanation text screen, as displayed during the adjustment procedure

On the text screen are, beside the possible commands, the measured values displayed. These measured values are constantly update during the adjustment. The 'bold' printed line in the example is the line with the important values. The purpose of the adjustment is to get these three values below 0.10.

Example:

Scan Util Scan Util : Settling Adjustment

dd-mmm-yyy

----- GCA: Gramp Settling Adjustment -----

POSSIBLE COMMANDS:

MASTER: Selects the Master channels SLAVE: Selects the Slave channels
 X/Y/Z: Selects channel for adjustment
 EXIT: Stops measurement and returns measured output parameters

MEASURED VALUES:

Adjusting channel: X (Master) Slope-End: 0.30 [ms] Current: 412.5 [A]

Found mx. deviations at	:	>100		>150		>250		[usec]
- Small spec. (w.r.t. fit)	:	0.68		0.23		0.04		[%]
- Wide spec. (w.r.t. 100%)	:	1.08		0.35		0.07		[%]

ENTER A COMMAND:

Possibilities: X,Y,Z,MASTER,SLAVE,EXIT
 Enter user input: _____

Explanation measured values display:

- Small spec. line: The values of the differences between the reference gradient response (on screen called ref. fit) and actual gradient response are displayed. The displayed values are measured at 100 usec, 150 msec and 250 usec after the end of the slope. All values must be adjusted to: < 0.10 %.
- Wide spec. line: The values of the differences between the demand pulse (on screen called ref. 100%) and the actual gradient response are displayed. Values are displayed at 100 usec, 150 usec and 250 usec after the end of the slope.
- Current: During the adjustment the gradient current is automatically set to 412.5 A and during the final measurement the gradient current is automatically set to 550 A.

SETTLING ADJUSTMENT SET UP FOR POWER/MASTER.

The settling procedure monitors the gradient current and the difference between the actual measured gradient current value and a reference gradient current curve.

Procedure:

1. Enable a monitor display in the technical room, which enables you to read the screen.
 See Section 11 (Extended Monitoring Appendix) how to do this.

WARNING

The noise level in front of the Copley-274 is very high. Hearing protection must be used.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

2. Login under GYROSCAN.

Select: <Scan Control>

Select: <Scan Utilities>

Select: Enter service mode <password>

Select: System tuning <password>

Select: System tuning tools

Select: Installation Procedures

Select: Gradient adjustment procedures

Select: Settling Adjustment

An information screen is displayed. Follow the screen messages.

Select: <Proceed> To start the measurement.

Two signals are displayed, the real gradient response (upper half (= signal 1)) and the difference signal (difference signal = reference gradient response - real gradient response) (lower half (= signal 2)).
(see figure 18 for POWER and figure 21 for MASTER)

3 Select: <Display grad>

4. Select: <Mod display>

Change/check:

monitoring mode both

Raw data file logging no

identifier of signal 1 Tr

sig 1: horizontal min 0

sig 1: horizontal max 2.3

sig 1: vertical min 98.0

sig 1: vertical max 101.0

sig 1: marker line no

Changing sig. 2 settings here is not possible.**(lines are overwritten by the test results from the running test)**

Select: <Proceed>

To zoom out a part of the signal 2 display, drag the desired window by moving the mouse, with the left mouse button pressed, over signal part of interest, or just toggle the autoscale button (on and off again).
By clicking the reset button, your original signal will return.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

2.4.1. SETTLING ADJUSTMENT PROCEDURE POWER

5. Enter: X
 6. Enter: master
 - Initial settings of the potentiometers: (all are 20 turns potentiometers, no end stop)
 - Turn the potentiometer 'DLY1' fully counter clockwise.
 - Turn the potentiometer 'HFEQ1' fully clockwise and 10 turns counter clockwise. (middle position)
 - Turn the potentiometer 'TC1' fully clockwise.
 - Turn the potentiometer 'BW1' fully clockwise.
- (see figure 16)

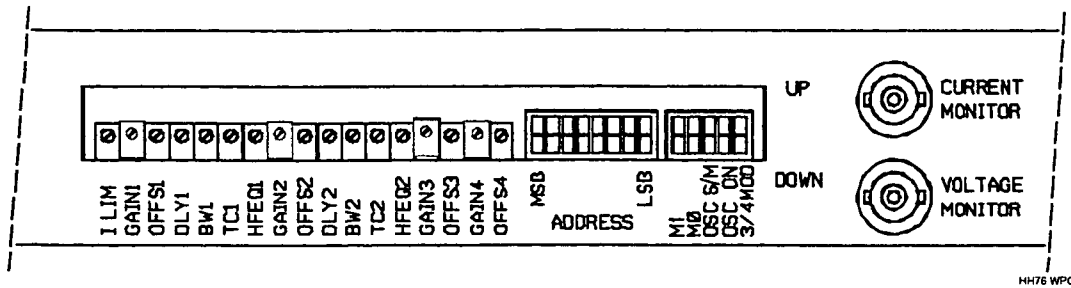


Figure 16: 'HFEQ1', 'TC1' and 'BW1' potentiometers location Copley-274 Controller

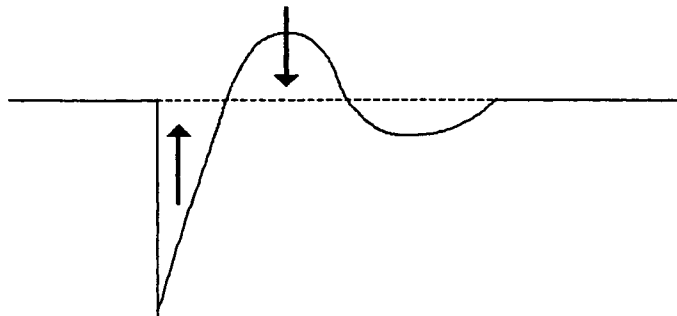


Figure 17: Adjustment principle

- Adjust the difference signal within specification, with potentiometers 'TC1' and 'BW1' on the X-master Copley-274 Controller, see figure 17 for the adjustment principle.
- First turn 'TC1' potentiometer until the one part of the signal is almost 0%, the other part of the signal will increase. Then turn 'BW1' potentiometer until this other part of the signal is almost 0%, the 'TC1' effected part of the signal will increase, but will stay below the starting value. Repeat the adjustment of the 'TC1' and 'BW1' potentiometers, until both signal parts are smaller then 0.10%. Check the text part of the screen for the actual values. Values are displayed in the line starting with '-Small spec.'.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

- Now you have to check the 'real' signal (see the upper signal in figure 18). Overshoots above the 100% level near the end of the slope, can shutdown the gradient amplifier, due to signal errors which result in a scan abort. A too slow response can cause gradient strength performance problems and can also shutdown the gradient amplifier, due to signal errors which result in a scan abort.
You cannot see this part of the signal in the difference signal, because the software does not display the slope information in the difference signal.
Use the HFEQ1 potentiometer to adjust the real signal according to the upper signal in figure 18 and keep the difference signal within spec with the BW1 and TC1 potentiometers.

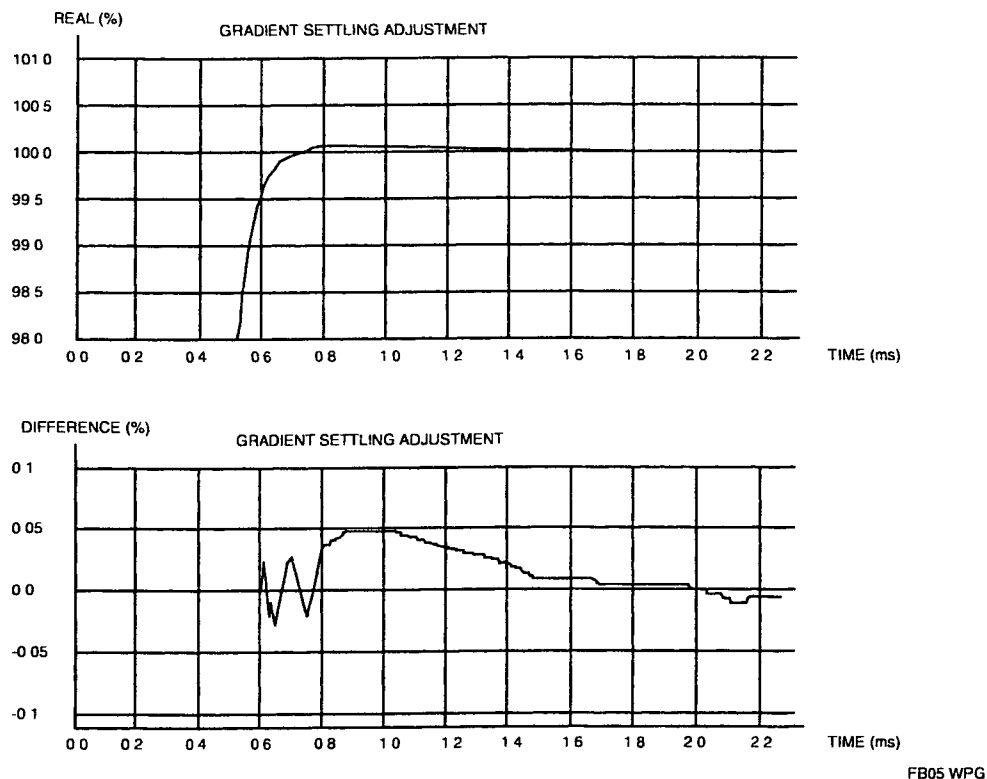


Figure 18: monitoring screen

Follow the same procedure and adjust:

- the Y master Copley-274 Controller potentiometers.
- the Z master Copley-274 Controller potentiometers.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

When all channels are adjusted:

9. Enter: Exit

The final settling adjustment check will check each channel successively.

In case of an error message, the operator must enter: <RETURN>. The program will continue the check until the next error message or all channels checked

The program is ready, when the 'gradient adjustment procedures' menu is displayed again.

10. If no error messages were seen, then the adjustment was done successfully. A MRL is available.
11. If an error message was seen, then repeat the adjustment for the failing channel(s). The experience has taught that after the second time adjustment all values are within the specifications.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

2.4.2. SETTLING ADJUSTMENT PROCEDURE MASTER

5. Enter: X

6. Enter: master

- The adjustments are done on the X master Copley-274 Controller.
- Initial settings of the potentiometers: (all are 20 turns potentiometers, no end stop)
- Turn the potentiometer 'DLY2' fully counter clockwise .
- Turn the potentiometer 'HFEQ2' fully clockwise and 10 turns counter clockwise (middle position)
- Turn the potentiometer 'TC2' fully clockwise
- Turn the potentiometer 'BW2' fully clockwise

(see figure 19)

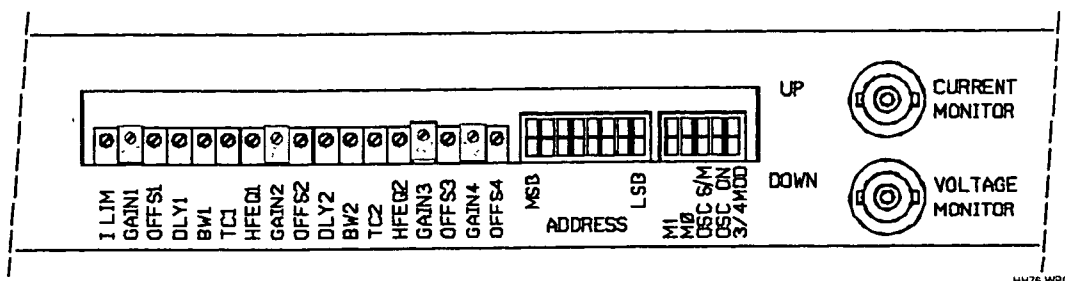


Figure 19: 'HFEQ2', 'TC2' and 'BW2' potentiometers location Copley-274 Controller

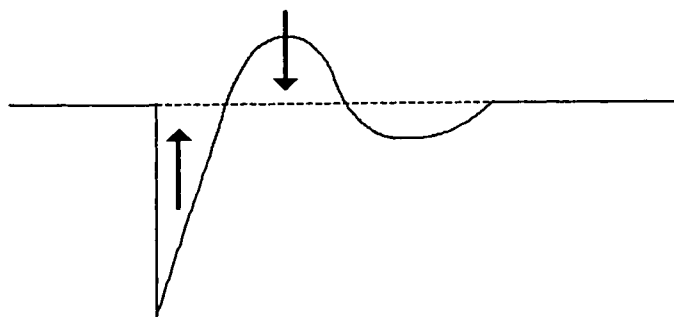


Figure 20: Adjustment principle

- Adjust the difference signal within specification, with potentiometers 'TC2' and 'BW2' on the X-master Copley-274 Controller, see figure 20 for the adjustment principle.
- First turn 'TC2' potentiometer until the one part of the signal is almost 0%, the other part of the signal will increase. Then turn 'BW2' potentiometer until this other part of the signal is almost 0%, the 'TC2' effected part of the signal will increase, but will stay below the starting value. Repeat the adjustment of the 'TC2' and 'BW2' potentiometers, until both signal parts are smaller then 0.10%. Check the text part of the screen for the actual values. Values are displayed in the line starting with
'-Small spec.'

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

- Now you have to check the 'real' signal. (see the upper signal in figure 21)
Overshoots above the 100% level near the end of the slope, can shutdown the gradient amplifier, due to signal errors which result in a scan abort. A too slow response can cause gradient strength performance problems and can also shutdown the gradient amplifier, due to signal errors which result in a scan abort.
You cannot see this part of the signal in the difference signal, because the software does not display the slope information in the difference signal.
Use the HFEQ2 potentiometer to adjust the real signal according to the upper signal in figure 21 and keep the difference signal within spec with the BW2 and TC2 potentiometers.

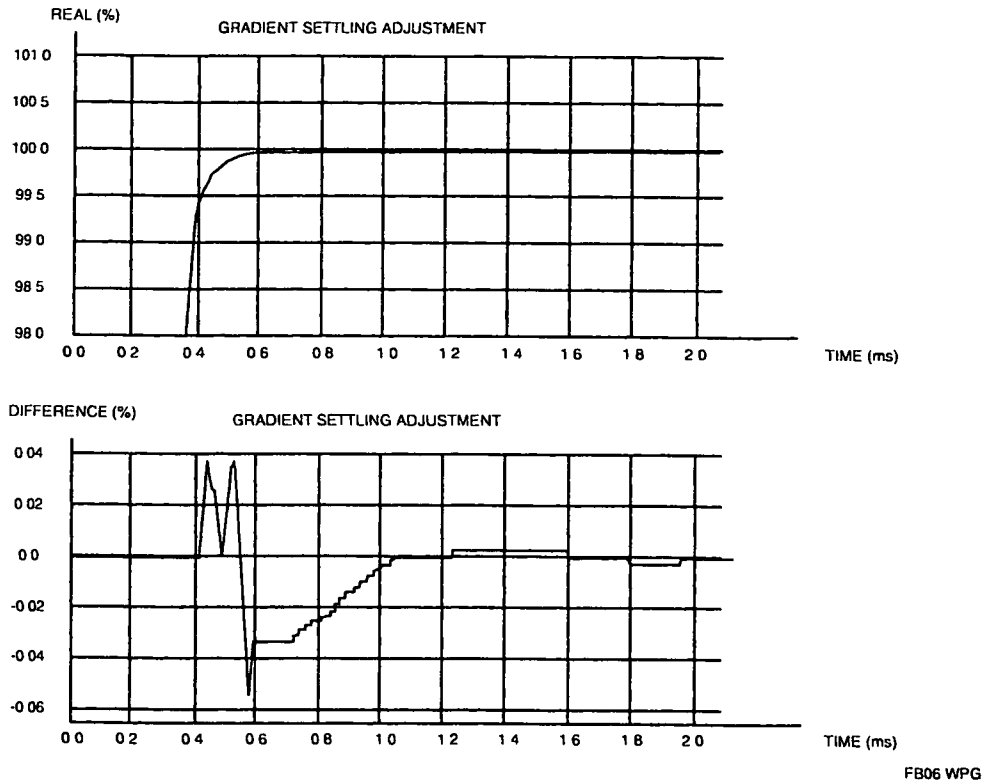


Figure 21 monitoring screen

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

7. Enter: slave

- Initial settings of the potentiometers: (all are 20 turns potentiometers, no end stop)
- Turn the potentiometer 'DLY2' fully counter clockwise .
- Turn the potentiometer 'HFEQ2' fully clockwise and 10 turns counter clockwise. (middle position)
- Turn the potentiometer 'TC2' fully clockwise.
- Turn the potentiometer 'BW2' fully clockwise.
- Adjust the difference signal and real signal, with potentiometers 'TC2' , 'BW2' and 'HFEQ2' on the X-slave Copley-274 Controller, as done on the master.

Follow the same procedure and adjust:

8. the Y master Copley-274 Controller potentiometers.
9. the Y slave Copley-274 Controller potentiometers.
10. the Z master Copley-274 Controller potentiometers.
11. the Z slave Copley-274 Controller potentiometers.

When all channels are adjusted:

12. Enter: Exit

The final settling adjustment check will check each channel successively.

In case of an error message, the operator must enter: <RETURN>. The program will continue the check until the next error message or all channels are checked.

The program is ready, when the 'gradient adjustment procedures' menu is displayed again.

13. If no error messages were seen, then the adjustment was done successfully. A MRL is available.
14. If an error message was seen, then repeat the adjustment for the failing channel(s). The experience has taught that after the second time adjustment all values are within the specifications.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

3. DUAL DELAY ADJUSTMENT (MASTER ONLY)

3.1. INTRODUCTION

The purpose of the 'dual delay adjustment' is to match the output of slave cabinet to the output of the master cabinet. This is done by measuring the difference in output-current using a dedicated Dual Mode Adjustment Tool (DMAT). The current-difference can be minimised by adjusting the gain, time-constant 'TC2' and bandwidth 'BW2' of the slave cabinet and the delay potentiometer 'DLY2' of the master or the slave cabinet.

Pre-conditions for this adjustment are:

- the DAC linearity correction must be carried out.
- the Gradient amplifier offset must be verified.
- the Gradient current has been adjusted.
- the Gradient settling behaviour has been adjusted.
- the Gradient orientation must be correct. The verification of the gradient orientation is described in section 16 of this Service Manual Installation (SMI).

Tools:

- Dual Mode Adjustment Tool (DMAT), code number: 8122 102 6441.

Offset adjustment procedure DMAT:

1. Connect the mains supply of the current probe to the mains outlet at the rear of Gradient Interface rack.
2. Put the sensitivity switch into position 200mV/A.
3. Make sure that the 'gradient' cables, coming out of the current probe, are 'open'. This means the cable ends do not touch each other.
4. Measure with a suitable digital voltmeter (DVM) on the output cable. Measure between the BNC connector housing and the BNC centre pin.
5. Adjust the potentiometer, located on top of the DMAT, until the offset value of the probe is 0 mV +/- 2 mV, sensitivity set to 200 mV/A.

3.2. MEASUREMENT SET-UP

WARNING

Be aware of the high voltage (420 Volt) present inside the gradient amplifier cabinet(s), on the connections to the gradient coil and the system filter box. Make sure that the AC power has been switched off for at least 2 minutes before starting to work on the gradient chain. It is advised to watch the high voltage value on the EMI 20kW/420V DC power supplies meters and make sure that the supply capacitors have discharged to less than 2 Volt.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

- 1 Enable a monitor display in the technical room, which enables you to read the screen.
See Section 11 how to do this

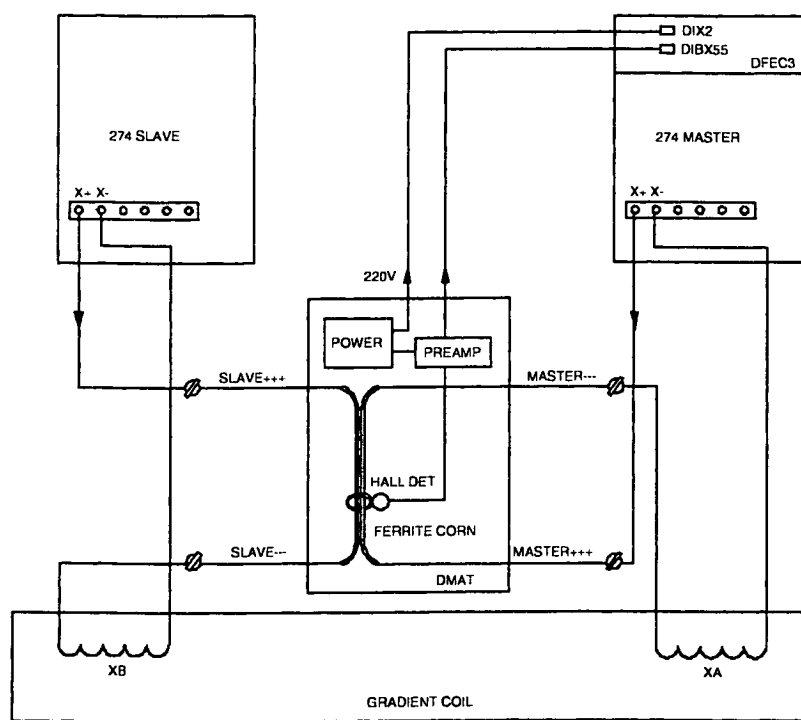
WARNING

The noise level in front of the Copley-274 is very high. Hearing protection must be used.

2. In the MDU, the gradient amplifiers should be switched off.

WARNING

*Be aware of the high voltage (420V) present on the external cables of the current probe device.
Make sure that all connections are insulated before switching on the gradient amplifier.*



HH74 WPG

Figure 22: Dual Mode Adjustment Tool (DMAT) connections

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

3. Connect the current probe device in the X chain according to figure 20.
 - a. Disconnect the gradient cables from X+ of the master gradient amplifier
 - b. Cable, marked with green label **master+++**, to X+ in the master gradient amplifier.
 - c. Shift the delivered rubber sleeve over the cable, marked with green label **master---**. Connect this cable to the gradient cables, which leads to x-coil (cables are disconnected in point a).
 - d. Shift the sleeve across the just made connection, this to prevent shorts to the cabinet.
 - e. Put the test cables in such a way that the rear door of the master gradient amplifier cabinet can be closed. Close the rear door of the master gradient amplifier.
 - f. Disconnect the gradient cables from X+ of the slave gradient amplifier
 - g. Cable, marked with blue label **slave+++**, to X+ in the slave gradient amplifier.
 - h. Shift the delivered rubber sleeve over the cable, marked with blue label **slave---**. Connect this cable to the gradient cables, which leads to x-coil (cables are disconnected in point f).
 - i. Shift the sleeve across the just made connection, this to prevent shorts to the cabinet.
 - j. Put the test cables in such a way that the rear door of the slave gradient amplifier cabinet can be closed. Close the rear door of the slave gradient amplifier.
 - k. Connect the DMAT output BNC cable to DFEC3, DIBX55.
 - l. Connect the DMAT mains cable to DFEC3, DIX2.
4. In the MDE switch on the gradient amplifier.
The gradient amplifiers cabinets are enabled after about 10 seconds.
5. Startup the dual delay adjustment:
Login under GYROSCAN.
 Select: <Scan Control>
 Select: <Scan Utilities>
 Select: Enter service mode <password>
 Select: System tuning <password>
 Select: System tuning tools
 Select: Installation Procedures
 Select: Gradient adjustment procedures
 Select: Dual delay adjustment
 An information screen is displayed.
 Select: <Proceed>
 Change/check:
 GCA: channel X

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

3.3. PRE-ADJUSTMENT CHECK

Before starting to re-adjust the amplifier, check if adjustment is required. When all specifications are met no adjustments are required.

Procedure:

1. Put the sensitivity_switch of the current probe device in position 50 mV/A.
2. Select: <Proceed> To start the measurement.
3. Switch on the grid in the Monitor display, if not already done.
4. Compare the measured signal with figure 22c.

The signal measured at 0.5, 3.0 and 5.5 msec should have the same amplitude. Normally there may be some small differences visible (= gain mismatch).

Specification: gain mismatch <10 mV (Sensitivity switch: 50 mV/A)

Some peaks will be visible. These peaks have an typical amplitude < 60 mV and should be symmetrical around the baseline.

Specification: slope1_integr. $0 \pm 6 \mu\text{Vs}$ (Sensitivity switch: 50 mV/A)
 slope2_integr. $0 \pm 6 \mu\text{Vs}$ (Sensitivity switch: 50 mV/A)
 max. peak in slope < 95 mV (Sensitivity switch: 50 mV/A)

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

3.4. ADJUSTMENT

- Put the sensitivity_switch of the current probe device in position 100 mV/A, the gain mismatch (=amplitude at 0.5, 3.0 and 5.5 ms) should be visible.

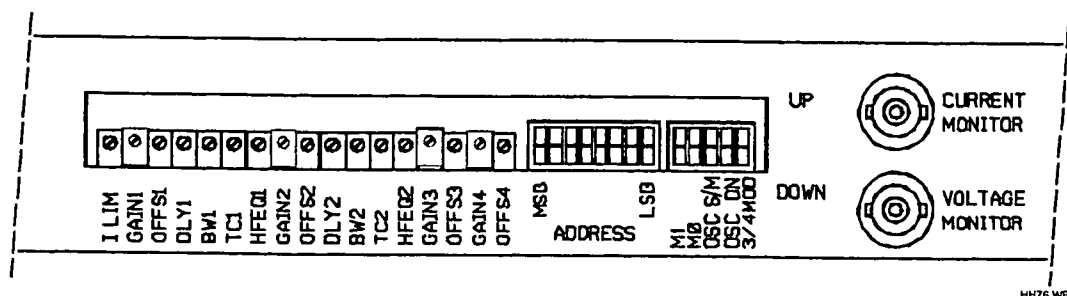
NOTE

Don't use automatic display scaling. Use the 'sensitivity' switch on the current probe to change the amplitudes of the monitor signal.

- Adjust the slave 'GAIN2' potentiometer in such way that the signal measured at 0.5, 3.0 and 5.5 msec have the same amplitude. Gain mismatch is minimal. Increase the sensitivity of the current probe, see figures 22a and 22b and continue the adjustment. Repeat the adjustment until a sensitivity of 50 mV/A has been reached and the gain mismatch is:
< 10 mV. (Sensitivity switch: 50 mV/A)
- Adjust the slope1 integr. and the max. peak values.
Try to reduce the slope1 integr. (is area of the pulse). Make sure that the signal is symmetrically around the base-line. Differences are caused by the settling behaviour of the slave.
Start the adjustment with the probe sensitivity_switch in a position which shows the complete pulse (10 mV/A till 50 mV/A). During the adjustment, the sensitivity can be increased.
Final adjustment must be done with the switch in position 50 mV/A. Refer to figure 22c for the final result.
The adjustment is done with the 'BW2', 'TC2' and 'HFEQ2' Slave Controller potentiometers
If the signal can not be optimised confirm figure 22c, then try to optimise the signal by turning the master- or slave delay 'DLY2' potentiometer slightly. Typical peak approx. 60 mV (Sensitivity switch: 50 mV/A)

<u>Specification:</u>	gain mismatch	<10 mV	(Sensitivity switch: 50 mV/A)
	slope1_integr.	$0 \pm 6 \mu\text{Vs}$	(Sensitivity switch: 50 mV/A)
	slope2_integr.	$0 \pm 6 \mu\text{Vs}$	(Sensitivity switch: 50 mV/A)
	max. peak in slope	< 95 mV	(Sensitivity switch: 50 mV/A)

- If the parameters 'gain mismatch', 'slope1_integr.', 'slope2_integr.' and 'max. peak in slope' are within the specification:
Enter: EXIT To stop the measurement.
- In the MDE switch off the gradient amplifier.
- Disconnect the current probe and re-connect the gradient coil cables to the terminal-block.
- Repeat the adjustment for the Y and Z gradient.



Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

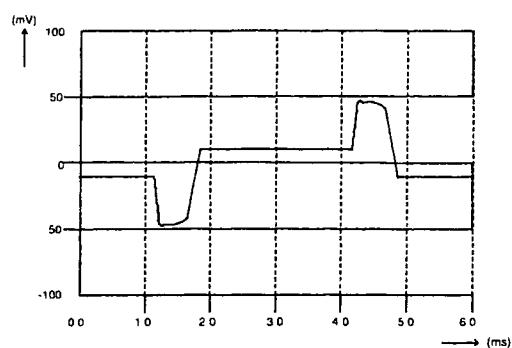


Figure 22A: Monitor signal before the 'GAIN2' adjustment

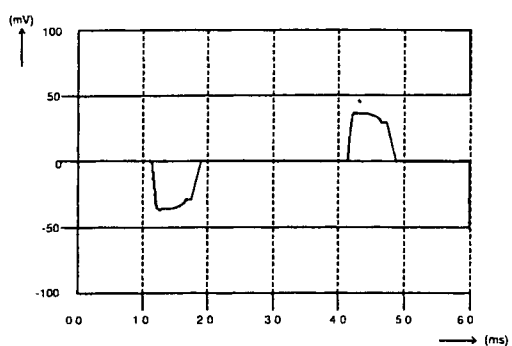


Figure 22B: Monitor signal after 'GAIN2' adjustment

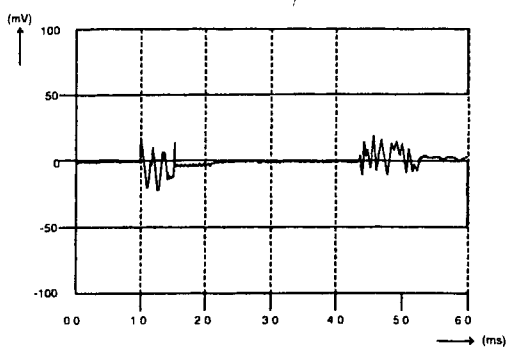


Figure 22C: Final monitor signal, current probe sensitivity set to 50mV/A

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

4. EDDY CURRENT CHECK PROCEDURE

Precondition:

- The gradient orientation must be correct, described in paragraph 'Check of the gradient orientation' of section 16 of this document

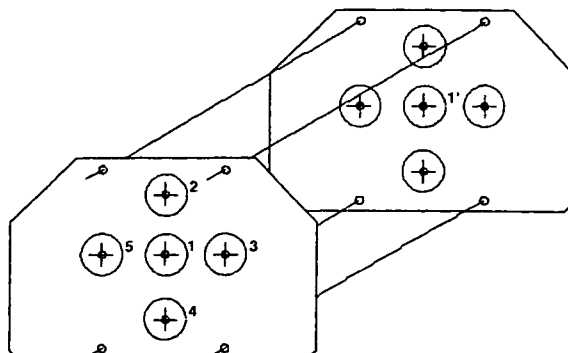
Required tools:

- Eddy current compensation set 8122 101 45004, which includes the Eddy current pick-up coils holder.
- The eddy currents are adjusted on the WFG/ECC boards inside the Gradient Interface rack. Since the measured signal is displayed on the operator console, a temporary display must be organised. For external monitoring in the technical room, see the appendix of chapter 11.

NOTE

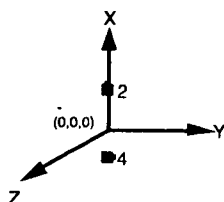
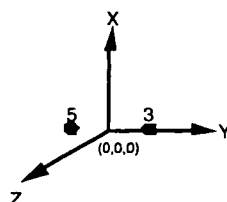
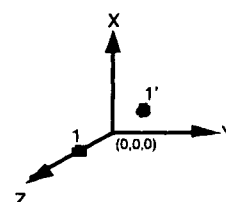
- *Take special care with positioning the fluxmeter, the drift is strongly dependent on environmental temperature variations.*
- *Keep the flux meter away from the gradient cabinet and gradient cables to eliminate the possibility of forced induction into the test equipment, the minimum distance > 1 meter.*
- *Use the coax cables, as delivered with the set, for the connection between the flux meter and the Gradient interface rack DIB to minimize the noise:*
 - *The 'thick' coax cables for the signals;*
 - *The 'thin' coax cable for the 'reset' signal.*

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master



JO19 WPG

Figure 23 - Position Eddy current pick-up coils

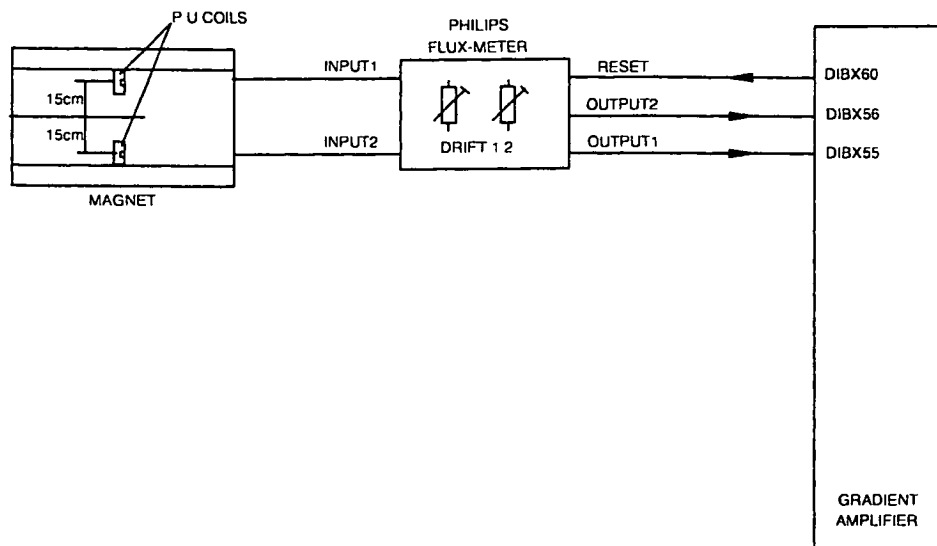
SET UP FOR FLUX DRIFT AND
X-WFG BOARD ADJUSTMENTSSET UP
Y-WFG BOARD ADJUSTMENTSSET UP
Z-WFG BOARD ADJUSTMENTS

JO19A WPG

Figure 24 - Position pick-up coils related to X,Y and Z gradient axis.

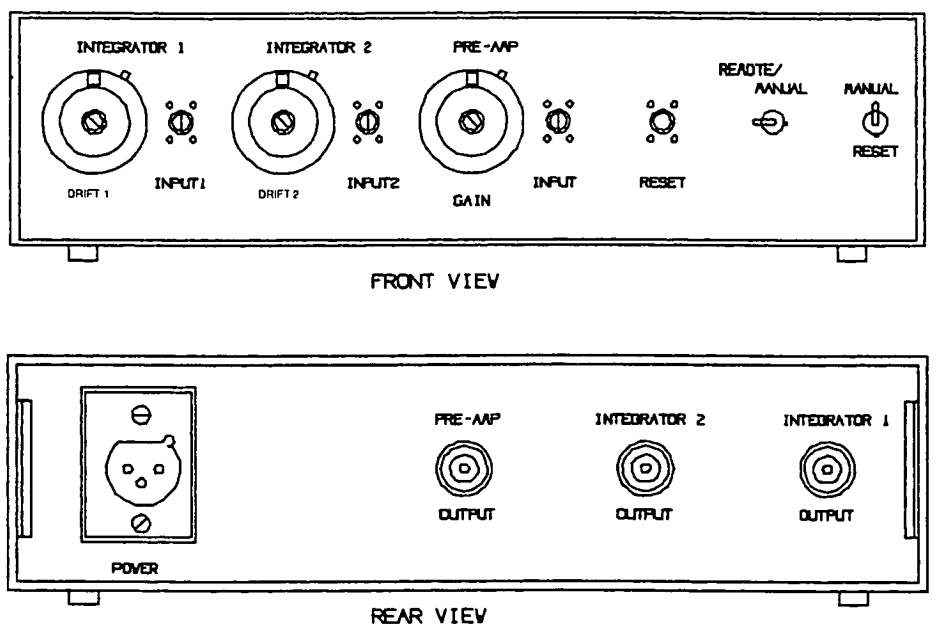
1. Mount the pick-up coils in the 15 cm positions for the 'gradient under test', see figure 23 and 24. 'Position Eddy current pick-up coils' and place the pick-up coils close to the iso centre of the bore, facing both the same side.
The axis of the coils must be always parallel to the Z direction. (accuracy +/- 5 degrees)
2. Switch on the fluxmeter and wait 30 minutes to stabilise.
3. Do the measurement set-up, see figure 25: 'Measuring set-up for Eddy current check' and figure 26: 'Front and rear view of Philips flux meter'.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master



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Figure 25: Measuring set-up for Eddy current check



EDDY.DPF

Figure 26: Front and rear view of Philips flux meter

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

4.1. DRIFT COMPENSATION FLUX METRE

Pre-condition Place the pick-up coils holder, with the pick-up coils in the direction, used for the gradient to be tested, on the pickup coils holder, close to the iso centre direct on the bridge section.

Procedure:

1. Set the flux meter remote/manual switch to remote.
2. The flux meter has to be switched **ON** for **at least 30 minutes**, before adjusting the offset (drift) of the flux meter.
3. Login under GYROSCAN.
 - Select: <Scan Control>
 - Select: <Scan Utilities>
 - Select: Enter service mode <password>
 - Select: System tuning <password>
 - Select: System tuning tools
 - Select: Installation procedures
 - Select: Gradient adjustment procedures
 - Select: Drift compensation flux metre.

An information screen is displayed.
4. Select: <Proceed> To start the measurement.
5. Select: <Mod display>
 - Change/check:
 - monitoring mode both
 - Raw data file logging no
 - identifier of signal 1 Tr
 - sig 1: horizontal min 1.0
 - sig 1: horizontal max 4990
 - sig 1: vertical min -0.1
 - sig 1: vertical max 0.1
 - sig 1: marker line no
 - identifier of signal 2 Ti
 - sig 2: horizontal min 1.0
 - sig 2: horizontal max 4990
 - sig 2: vertical min -0.1
 - sig 2: vertical max 0.1
 - sig 2: marker line no
 - Select: <Proceed>
6. Select: <Display grid>
7. Select: <Display min/max>
8. Adjust, on the fluxmeter both potentiometers DRIFT 1 and 2, the drift of the both channels until the signal stays within **the limit of +/- 10 mV**. (Drift is the difference between the voltage on 1 ms and 5 seconds). To see the effect of the adjustment, one must wait for at least two pulses, approx. 30s.
9. Select: <Stop scan> To stop the measurement.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

4.2. POSITIONING EDDY CURRENT PICK-UP COILS

Pre-condition: Place the pick-up coils holder, with the pick-up coils in 'xxx'-direction on the pickup coils holder, close to the iso centre.
'xxx' stands for gradient under test.

1. Select: Position eddy current pickup coils
An information screen is displayed.
2. Select: <Proceed>
3. Change/check: gradient coil Z
4. Select: <Proceed> To start the measurement.
5. If pickup coils are positioned for the X- or Y-gradient adjustment:
Select: <Mod display>
Change/check:

monitoring mode	both
Raw data file logging	no
identifier of signal 1	Tr
sig 1: horizontal min	1.0
sig 1: horizontal max	150
sig 1: vertical min	-0.1
sig 1: vertical max	0.1
sig 1: marker line	no
identifier of signal 2	Ti
sig 2: horizontal min	1.0
sig 2: horizontal max	150
sig 2: vertical min	-0.1
sig 2: vertical max	0.1
sig 2: marker line	no

Select: <Proceed>
6. If pickup coils are positioned for the X- or Y-gradient adjustment:
Move the holder with the pick-up coils in Z-direction, until the response of the two channels is:
- 50 mV < V < + 50 mV.
If pickup coils are positioned for the Z-gradient adjustment:
Move the holder with the pick-up coils in Z-direction, until the response of the two channels is:
two signals with the same amplitude (absolute).
7. Select: <Stop scan>
8. To reduce the positioning time for the other gradients, mark the position of the pickup coils holder.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

4.3. EDDY CURRENT CHECK

Since the Eddy currents have been adjusted in the factory, it is advised to check the actual behaviour before starting to make any readjustments. For this check, the 'eddy current final measurement' procedure will be used. During the final measurement the specific channel delay deviation is changed **till the three quarter point specification is reached**. The flatness of the averaged values are measured for the specified ranges. After completion of the final measurement the measured parameters will be checked against the specifications. A MRL will be displayed. If all values are within the limits the values are stored automatically into the tuned hardware parameter archive.

Ensure you are in the 'Gradient adjustment procedures ' menu

1. Select. Eddy current final measurement
An information screen is displayed.
2. Select. <Proceed>
3. Change/check: gradient coil xxx
'xxx' stands for gradient under test.
4. Change/check: Current ECC settings
Activate all filters:

1	ON	2	ON	3	ON
4	ON	5	ON	6	ON
5. Select: <Proceed> To start the measurement.

At the end of measurement a measurement result list (MRL) will be displayed

Specifications:

		POWER	MASTER	Specification
ECC:	Three-quarter point	0.30 ms		69.0 % $\pm$ 3.0 %
ECC:	Three-quarter point		0.15 ms	63.0 % $\pm$ 6.0 %
ECC:	EOS point	0.4 ms		95.0 % $\pm$ 0.5 %
ECC:	EOS point		0.224 ms	91.5 % $\pm$ 0.5 %
ECC:	Flatness 1	0.7 -> 3.4 ms	0.5 ms -> 3.2 ms	< 0.15 % *)
ECC:	Flatness 2	3.4 -> 990 ms	3.3 ms -> 990 ms	< 0.1 % *)
ECC:	Difference 1	1.0 -> 5.4 ms	0.8 -> 5.2 ms	< 0.45 % **)
ECC:	Difference 2	5.8 -> 990 ms	5.2 -> 990 ms	< 0.42 % **)

*) $[(\text{max.value}) - (\text{min.value})]$

```
**) [(max.value) - (min.value)]
```

- If the system does not pass the **flatness** tests continue with paragraph 5 'Eddy current adjustment procedure'. Otherwise change the position of the pickup coils for another axis. Repeat the pickup coils positioning and start the final measurement again.
- If the system does not pass the **difference** tests, then the most likely cause is the fluxmeter. To exclude the measurement tool, exchange the eddy current probe connections at the front of the fluxmeter. Check the final measurement again. If difference still not within the specification contact the MR helpdesk. In the mean time change the position of the pickup coils for another axis. Repeat the pickup coils positioning and start the final measurement again.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

SUMMARY of Eddy current compensation check:**X-axis**

- Put the two pick-up coils to position 2 and 4 on the X-axis, see figure 23 'Position Eddy current pick-up coils'.
Do the 'Positioning eddy current pickup coils'.
Select the Z gradient and move the holder with the pick-up coils in Z-direction, until the response of the two channels is $< \pm 50$ mV.
- Do the 'Eddy current final measurement' for the X gradient.

Y-axis

- Put the two pick-up coils to position 3 and 5 on the Y-axis, see figure 23 'Position Eddy current pick-up coils'.
Do the 'Positioning eddy current pickup coils'.
Select the Z gradient and move the holder with the pick-up coils in Z-direction, until the response of the two channels is $< \pm 50$ mV.
- Do the 'Eddy current final measurement' for the Y gradient.

Z-axis

- Put the two pick-up coils to position 1 and 1' on the Z-axis, see figure 23 'Position Eddy current pick-up coils'.
Do the 'Positioning eddy current pickup coils'.
Select the Z gradient and move the holder with the pick-up coils in Z-direction, until the response of the two channels are equal, tolerance ± 100 mV.
- Do the 'Eddy current final measurement' for the Z gradient.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

5. EDDY CURRENT ADJUSTMENT PROCEDURE

5.1. INTRODUCTION

Adjustment of the magnetic field response can be done with 6 correction filters. A seventh filter A0/T0, adjusted and sealed in the factory, is not used in this procedure. The potentiometers to adjust time constant and gain factor are positioned on the WFG/ECC boards, which are located at the interface unit (DI) in the gradient amplifier cabinet. The six filters can be activated by ASW. In the Hardware parameter administration is stored which filter is active or not active. If active, the corresponding **red** led on the WFG board is on during SCAN. The filters are set during the Eddy current adjustment procedure and later on stored <STORE>.

Changing T(x) will result in another time constant and changing A(x) will lead to changing the gain factor/sign within the range of the T(x) time constant (positive or negative compensation). In the middle of the 15 turns gain potentiometer the gain is zero. By turning clock wise (CW) it gives more and more positive compensation and by turning counter clock wise (CCW) just the reverse (negative). The next table shows the different operation ranges.

Filter	Switch	Operation range (msec)	
A6/T6	Filter 6	200	3600
A5/T5	Filter 5	40	640
A4/T4	Filter 4	4	64
A3/T3	Filter 3	0.4	6.4
A2/T2	Filter 2	0.2	2.2
A1/T1	Filter 1	0.03	0.33

A filter setting (A,T) must be adjusted in case shape part A of figure 27 is found, i.e. an overshoot followed by an undershoot. If this shape is found, both potentiometers must be adjusted iterative T(x) (CW) and A counter (CCW) to reduce the overshoot and undershoot. But be aware never loose the principle shape overshoot followed by an undershoot.

Start with turning the gain potentiometer (A) of the filter to be adjusted clockwise till a clear overshoot can be seen.

WARNING

The noise level in front of the Copley-274 is very high. Hearing protection must be used.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Depending on the shape of the overshoot a certain correction procedure must be chosen.

In part A of figure 27, the time constant is too small, increase T time constant potentiometer.

In part B of figure 27, the time constant is too large and/or the amplitude too high. Change both potentiometers to correct it.

FIGURE A FIELD RESPONSE TIME
CONSTANT TOO HIGH

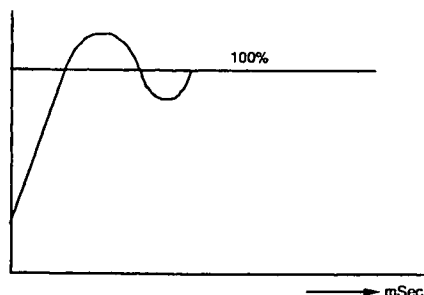
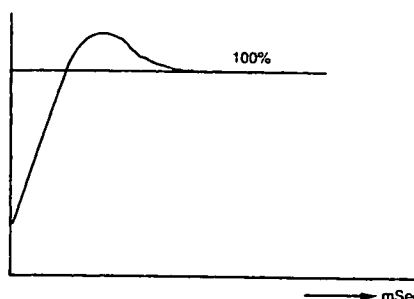


FIGURE B FIELD RESPONSE TIME
CONSTANT TOO LARGE
AND/OR AMPLITUDE TOO HIGH



HH11 WPG

Figure 27: Field response as function of time.

Pre-condition:

- The pick-up coils are placed in direction under test in the marked position of the holder. Before starting the Eddy current correction procedure do a 'Reference measurement'. During this adjustment the gradient channel delay parameters, determined in the procedures 'Eddy current final measurement' and/or 'Gradient channel delay determination', will be set automatically to the default value 0.025 msec.

Start-up procedure for the following paragraphs

- Login under GYROSCAN.
 - Select: <Scan Control>
 - Select: <Scan Utilities>
 - Select: Enter service mode <password>
 - Select: System tuning <password>
 - Select: System tuning tools
 - Select: Installation procedures
 - Select: Gradient adjustment procedures

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

5.2. SETTING STARTING POINT FOR THE POTENTIOMETERS (OPTIONAL)

After installing a new board or in case the flatness of the eddy-current compensation is completely out of specification, it is advised to return the potentiometers to the starting position (= centre (zero) position). In general however it should be sufficient to start the adjustment without turning the potentiometers first.

Turning the potentiometers to the starting position.

Turn the 6 A's and 6 T's potentiometers of X,Y or Z WFG/ECC board fully CCW and after that turn all A's 7.5 turns CW. (gain zero). An overshoot can be expected in this starting condition

5.3. EDDY CURRENT REFERENCE MEASUREMENT

Pre-condition: The pick-up coils are placed in the direction under test.

Procedure:

- 1 Select: Eddy current reference measurement
An information screen is displayed.
- 2 Select: <Proceed>
- 3 Change/check:

Measuring type	Reference	
Gradient coil	Gradient under test	equals position of the pick-up coils
Gradient ampl. faults	enabled	
- 4 Select: <Proceed> to start the measurement.
The reference measurement is performed and the maximum gradient strength is measured.
The program will display the result

The maximum gradient strength should be: **> 4.5 mT/m**

- 5 Enter: <RETURN> to accept this condition.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

5.4. WFG/ECC EDDY CURRENT BOARD ADJUSTMENT PROCEDURE

NOTE

Do not leave any potentiometer in the centre (zero) position. This to avoid problems with the next (shorter) time constant.

Try to maximise the gain (A) and minimise the time-constant (T) settings.

1. Select: Compensate long term eddy current
2. Change/check gradient coil xxx
'xxx' stands for gradient under test.
Current ECC settings

1	OFF	2	OFF	3	OFF
4	OFF	5	OFF	6	ON

3. Select: <Proceed> To start the adjustment of filter 6.
4. Select: <Mod display>

Change/check:

monitoring mode	both
raw data logging	no
identifier of signal 1	Tavg
sig 1: horizontal min	5.0
sig 1: horizontal max	990
sig 1: vertical min	99.0
sig 1: vertical max	101.0
sig 1: marker line	no
identifier of signal 2	Tdif
sig 2: horizontal min	5.0
sig 2: horizontal max	990
sig 2: vertical min	-1.00
sig 2: vertical max	1.00
sig 2: marker line	no

5. Select: <Proceed>
6. Adjustment of the A6/T6 filter. Check the flatness. If the flatness is not correct try to improve it by tuning only the potentiometer A6 a little bit. Do not yet touch T6.
The flatness should be < 0.1 %, where the flatness is defined as: $\frac{[(\text{max.value}) - (\text{min.value})]}{(\text{max.value}) + (\text{min.value})}$
When you do not succeed with this adjustment then turn T6 a little bit clock wise and turn A6 clock wise (+GAIN) till the last part of the displayed curve is flat as possible, but at least within the specifications.
While turning A6 CW you are only able to lift-up the curve. If already a overshoot exists use the negative compensation turning A6 CCW (-GAIN). If within specifications, continue with the next point.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

In case an overshoot is followed by an undershoot, then:

- Turn T6 (+TIME) CW till the undershoot is reduced till 1/16 of the overshoot and turn A6 (-GAIN) CCW till the amplitude of the overshoot above the 100% level is equal to the amplitude of the undershoot under the 100% level. Repeat this action till the optimum result is reached.
- The best Eddy current compensation is reached by maximum gain (A) and minimum time (T) settings.
- If necessary modify the vertical scaling factor, select with mouse 'Mod Display'. This also applies to the other filter combinations.

7. Select: <Stop scan>

8. Change/check: Current ECC settings

1	OFF	2	OFF	3	OFF
4	OFF	5	ON	6	ON

9. Select: <Proceed> twice To start the adjustment of filter 5.

10. Adjustment of the A5/T5 filter. Check the flatness. If the flatness is not correct try to improve it by tuning only the potentiometer A5 a little bit. Do not yet touch T5.

The flatness should be < 0.1 %, where the flatness is defined as: $[(\text{max.value}) - (\text{min.value})]$

When you do not succeed with this adjustment then turn T5 a little bit clock wise and turn A5 clock wise (+GAIN) till the last part of the displayed curve is flat as possible, but at least within the specifications.

While turning A5 CW you are only able to lift-up the curve. If already a overshoot exists use the negative compensation turning A5 CCW (-GAIN). If within specifications, continue with the next point.

11. Select: <Stop scan>

12. Change/check: Current ECC settings

1	OFF	2	OFF	3	OFF
4	ON	5	ON	6	ON

13. Select: <Proceed> twice To start the adjustment of filter 4.

14. Adjustment of the A4/T4 filter. Check the flatness. If the flatness is not correct try to improve it by tuning only the potentiometer A4 a little bit. Do not yet touch T4.

The flatness should be < 0.1 %, where the flatness is defined as: $[(\text{max.value}) - (\text{min.value})]$

When you do not succeed with this adjustment then turn T4 a little bit clock wise and turn A4 clock wise (+GAIN) till the last part of the displayed curve is flat as possible, but at least within the specifications.

While turning A4 CW you are only able to lift-up the curve. If already a overshoot exists use the negative compensation turning A4 CCW (-GAIN). If within specifications, continue with the next point.

15. Select: <Stop scan>

16. Select: <Cancel>

17. Select: Compensate mid term eddy current

An information screen is displayed.

18. Select: <Proceed>

19. Change/check: Current ECC settings

1	OFF	2	OFF	3	ON
4	ON	5	ON	6	ON

20. Select: <Proceed> twice To start the adjustment of filter 3.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

21. Adjustment of the A3/T3 filter. Check the flatness. If the flatness is not correct try to improve it by tuning only the potentiometer A3 a little bit. Do not yet touch T3.

The flatness should be (where the flatness is defined as:

$[(\text{max.value}) - (\text{min.value})]$):

From 1 ms until 3.6 ms: **< 0.15 %**

From 3.6 ms onwards: **< 0.1 %**

When you do not succeed with this adjustment then turn T3 a little bit clock wise and turn A3 clock wise (+GAIN) till the last part of the displayed curve is flat as possible, but at least within the specifications.

While turning A3 CW you are only able to lift-up the curve. If already a overshoot exists use the negative compensation turning A3 CCW (-GAIN). If within specifications, continue with the next point.

22. Select: <Stop scan>
 23. Select: <Cancel>
 24. Select: Compensate short term eddy current
 25. Change/check. Current ECC settings

1	OFF	2	ON	3	ON
4	ON	5	ON	6	ON

26. Select: <Proceed> twice To start the adjustment of filter 2.

27. Select: <Mod display>

Change/check:

monitoring mode	one
raw data logging	no
identifier of signal 1	Tavg
sig 1: horizontal min	0.90
sig 1: horizontal max	5.00
sig 1: vertical min	99.8
sig 1: vertical max	100.2
sig 1: marker line	no

Select: <Proceed>

28. Adjustment of the A2/T2 filter. Check the flatness. If the flatness is not correct try to improve it by tuning only the potentiometer A2 a little bit. Do not yet touch T2.

The flatness should be (where the flatness is defined as:

$[(\text{max.value}) - (\text{min.value})]$):

POWER:

From 0.7 ms until 3.4 ms: **< 0.15 %**

From 3.4 ms onwards: **< 0.1 %**

MASTER:

From 0.5 ms until 3.2 ms: **< 0.15 %**

From 3.2 ms onwards: **< 0.1 %**

Above flatness range must be adjusted with A2 and T2. Because of the larger bandwidth of the Copley-274 this should not present a problem. If however a problem arises, this is probably caused by the settling adjustment.

When you do not succeed with this adjustment then turn T2 a little bit clock wise and turn A2 clock wise (+GAIN) till the displayed curve is flat as possible, but at least within the specifications. While turning A2 CW you are only able to lift-up the curve. If already a overshoot exists use the negative compensation turning A2 CCW (-GAIN). If within specifications, continue with the next point.

29. Select: <Stop scan>

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

30. Change/check: Current ECC settings

1	ON	2	ON	3	ON
4	ON	5	ON	6	ON

31. Select: <Proceed> twice To start the adjustment of filter 1.

32. Select: <Mod display>

Change/check:

	POWER	MASTER
monitoring mode	one	one
raw data logging	no	no
identifier of signal 1	Tavg	Tavg
sig 1: horizontal min	0.70	0.50
sig 1: horizontal max	5.00	5.00
sig 1: vertical min	99.8	99.8
sig 1: vertical max	100.2	100.2
sig 1. marker line	no	no

Select: <Proceed>

33. Turn A1 and T1 potentiometers fully counter clockwise.

34. Turn T1 clockwise until its response 'reaches' the start of the flatness specification range. A small influence will be seen on the amplitude of A1. Correct this by adjusting A2 slightly. Next turn A1 potentiometer clockwise until the response at 3/4-point reaches its specifications. The response is displayed on the 'menu' selection window. Response value is coloured RED if not within specifications.

Remark for POWER only:

Every time the amplitude settings of A1 is changed an influence of the A2 amplitude can be seen. This means one must iterate between A1 and A2. Where A1 is used to get the marker value within the specifications (95.0% +/- 0.5%), and A2 to restore the fluidness adjustment. This procedure gets more difficult if one also changes T1 or T2 during the iteration. The trick is, first to set T1 and T2 correctly, and not to change it afterwards.

Remark for MASTER only:

Every time the amplitude settings of A1 is changed an influence of the A2 amplitude can be seen. This means one must iterate between A1 and A2. Where A1 is used to get the marker value within the specifications (91.5% +/- 0.5%), and A2 to restore the fluidness adjustment. This procedure gets more difficult if one also changes T1 or T2 during the iteration. The trick is, first to set T1 and T2 correctly, and not to change it afterwards.

The flatness should be (where the flatness is defined as:

[(max.value) - (min.value)]:

POWER:

From 0.7 ms until 3.4 ms: < 0.15 %

From 3.4 ms onwards: < 0.1 %

MASTER:

From 0.5 ms until 3.2 ms: < 0.15 %

From 3.2 ms onwards: < 0.1 %

Check also the response at 3/4 point:

POWER: 0.30 msec, should be: 69.0 % +/- 3.0 %.

MASTER: 0.15 msec, should be: 63.0 % +/- 6.0 %.

The response is displayed on the 'menu' selection window. Response value is coloured RED if not within specifications.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Remark for POWER only: Check also the response at EOS point:

Response at EOS point: 95.0 % +/- 0.5 %.

Remark for MASTER only: Check also the response at EOS point:

Response at EOS point: 91.5 % +/- 0.5 %

The response is displayed on the 'menu' selection window. Response value is coloured RED if not within specifications

35. Select: <Stop scan>

36. Select: <Cancel>

5.5. EDDY CURRENT FINAL MEASUREMENT

1. Select: Eddy current final measurement

An information screen is displayed.

2. Select: <Proceed>

3. Change/check: gradient coil xxx
'xxx' stands for gradient under test.

4. Change/check: Settings

1	ON	2	ON	3	ON
4	ON	5	ON	6	ON

5. Select: <Proceed> To start the measurement.

At the end of measurement a measurement result list (MRL) will be displayed.

For specifications see paragraph 4.3.

- If the system does not pass the **flatness** tests, check the times of the extreme values. Depending of these times a part of paragraph 5 'Eddy current adjustment procedure' must be repeated. Otherwise change the position of the pickup coils for another axis. Repeat the pickup coils positioning and start the final measurement again.
- If the system does not pass the **difference** tests, then the most likely cause is the fluxmeter. To exclude the measurement tool, exchange the eddy current probe connections at the front of the fluxmeter. Check the final measurement again. If difference still not within the specification contact the MR helpdesk. In the mean time change the position of the pickup coils for another axis. Repeat the pickup coils positioning and start the final measurement again.

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

6. EXAMPLES OF MRL PRINT OUTS

Example of MRL print out POWER, with some comments added.

Line Description

Line	Description	Final	Nominal	Spec
1.	measurement type	Y		
2.	gradient coil X,Y or Z	Y		
3.	current ECC settings Y 1	<value>		
4.	current ECC settings Y 2	<value>		
5.	current ECC settings Y 3	<value>		
6.	current ECC settings Y 4	<value>		
7.	current ECC settings Y 5	<value>		
8.	current ECC settings Y 6	<value>		
9.	settings Y 1	<value>		
10.	settings Y 2	<value>		
11.	settings Y 3	<value>		
12.	settings Y 4	<value>		
13.	settings Y 5	<value>		
14.	settings Y 6	<value>		
15.	overall gradient delay (msec)	<value>		
16.	delay deviation (us)	<value>	0.0	[-12.0 - 12.0]
17.	three-quarter point (0.30 msec)	<value>	69.00	[66.00 - 72.00]
18.	EOS point (%)	<value>	95.0	[94.5 - 95.5]
19.	Flatness range 1 (%)	<value>	0.00	< 0.15
20.	Flatness range 2 (%)	<value>	0.00	< 0.10
21.	Difference 1 (%)	<value>	0.00	< 0.45
22.	Difference 2 (%)	<value>	0.00	< 0.42
23.	Extreme times (ms) 1	<value>		
24.	Extreme times (ms) 2	<value>		
25.	Extreme times (ms) 3	<value>		
26.	Extreme times (ms) 4	<value>		
27.	Extreme times (ms) 5	<value>		
28.	Extreme times (ms) 6	<value>		
29.	Extreme times (ms) 7	<value>		
30.	Extreme times (ms) 8	<value>		
31.	Spec. ranges (ms) 1	0.30		
32.	Spec. ranges (ms) 2	3.00		
33.	Spec. ranges (ms) 3	3.00		
34.	Spec. ranges (ms) 4	990.00		
35.	Spec. ranges (ms) 5	0.60		
36.	Spec. ranges (ms) 6	5.00		
37.	Spec. ranges (ms) 7	5.00		
38.	Spec. ranges (ms) 8	990.00		
39.	Meas-system delay [us]	24.00	24.00	[23.50 - 24.50]

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Example of MRL print out MASTER, with some comments added:

Line	Description		Final	Nominal	Spec
1.	measurement type		Y		
2.	gradient coil X,Y or Z		Y		
3.	current ECC settings Y	1	<value>		
4.	current ECC settings Y	2	<value>		
5.	current ECC settings Y	3	<value>		
6.	current ECC settings Y	4	<value>		
7.	current ECC settings Y	5	<value>		
8.	current ECC settings Y	6	<value>		
9.	settings Y	1	<value>		
10.	settings Y	2	<value>		
11.	settings Y	3	<value>		
12.	settings Y	4	<value>		
13.	settings Y	5	<value>		
14.	settings Y	6	<value>		
15.	overall gradient delay (msec)		<value>		
16.	delay deviation (us)		<value>	0.0	[-12.0 - 12.0]
17.	three-quarter point (0.15 msec)		<value>	63.00	[57.00 - 69.00]
18.	EOS point (%)		<value>	91.5	[91.0 - 92.0]
19.	Flatness range 1 (%)		<value>	0.00	< 0.15
20.	Flatness range 2 (%)		<value>	0.00	< 0.10
21.	Difference 1 (%)		<value>	0.00	< 0.45
22.	Difference 2 (%)		<value>	0.00	< 0.42
23.	Extreme times (ms) 1		<value>		
24.	Extreme times (ms) 2		<value>		
25.	Extreme times (ms) 3		<value>		
26.	Extreme times (ms) 4		<value>		
27.	Extreme times (ms) 5		<value>		
28.	Extreme times (ms) 6		<value>		
29.	Extreme times (ms) 7		<value>		
30.	Extreme times (ms) 8		<value>		
31.	Spec. ranges (ms) 1		0.30		
32.	Spec. ranges (ms) 2		3.00		
33.	Spec. ranges (ms) 3		3.00		
34.	Spec. ranges (ms) 4		990.00		
35.	Spec. ranges (ms) 5		0.60		
36.	Spec. ranges (ms) 6		5.00		
37.	Spec. ranges (ms) 7		5.00		
38.	Spec. ranges (ms) 8		990.00		
39.	Meas-system delay [us]		24.00	24.00	[23.50 - 24.50]

Intera 0.5 T	Intera 1.0 T	Intera 1.5 T
	Power	Power Master

Remarks:

In the actual display of the result screen the line numbers are not shown

Line

Line 23	Time at which minimum of averaged signal is measured for flatness 1
Line 24	Time at which maximum of averaged signal is measured for flatness 1
Line 25	Time at which minimum of averaged signal is measured for flatness 2
Line 26	Time at which maximum of averaged signal is measured for flatness 2
Line 27	Time at which minimum of difference signal is measured for difference 1
Line 28	Time at which maximum of difference signal is measured for difference 1
Line 29	Time at which minimum of difference signal is measured for difference 2
Line 30	Time at which maximum of difference signal is measured for difference 2
Line 31	Start time specification range for flatness 1. POWER: add 0.4 ms to find actual start time MASTER: add 0.2 ms to find actual start time
Line 32	End time specification range for flatness 1. POWER: add 0.4 ms to find actual end time MASTER: add 0.2 ms to find actual end time
Line 33	Start time specification range for flatness 2. POWER: add 0.4 ms to find actual start time MASTER: add 0.2 ms to find actual start time
Line 34	End time specification range for flatness 2.
Line 35	Start time specification range for difference 1 POWER: add 0.4 ms to find actual start time MASTER: add 0.2 ms to find actual start time
Line 36	End time specification range for difference 1 POWER: add 0.4 ms to find actual start time MASTER: add 0.2 ms to find actual start time
Line 37	Start time specification range for difference 2 POWER: add 0.4 ms to find actual start time MASTER: add 0.2 ms to find actual start time
Line 38	End time specification range for flatness difference 2
Line 39	Meas-system delay